

Corridor, Not Factory: Trade Reorientation and the Missing Investment Response in Kazakhstan, 2022–2025

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Abstract

A large, sector-specific rise in demand for a tradeable good is expected to induce domestic investment. After February 2022, with Russia’s direct trade with its main partners disrupted, part of it was reallocated through neighbouring economies, Kazakhstan among them. Using product-level trade data (HS6, 2018–2025) we measure the demand shock, estimate what Kazakhstan retains, and ask whether it induced investment. Exports to Russia in 29 product lines break discontinuously in May 2022 and rise roughly tenfold (a selection-free, externally compiled priority list not chosen on this outcome rises $3.5\times$ over the same window); the break date survives an HAC correction, but the basket’s difference-in-differences is not separable from its own selection rule, and a 50-line externally compiled priority list corroborates the break date and, on its 26 non-overlapping lines against a purged control, a smaller-magnitude version of the effect, so identification rests on the break dates, raw levels, and a neighbour comparison rather than on the cross-sectional regression alone. Propagating a wholesale-and-freight margin (6–14%, a band we choose rather than estimate) through the input–output table implies domestic value added of 5–11 cents per rerouted dollar,

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against about three-quarters for a produced dollar, a reading that holds for any margin under about a fifth. Across three commercial deal databases we find no investment response in the sectors the flow passes through, though four post-2022 years rule out only a large one. We read the null through three gates — market access, irreversibility and capital-market institutions. An extreme irreversibility configuration or a capital market that cannot fund the projects that pass would each drive the response toward zero on their own; an open market-access gate does so only jointly with a small transformation margin, not by itself. The case identifies an open market-access gate, created by a customs union that makes trade a close substitute for production, together with suggestive (not dispositive) evidence for that small margin, and finds that the least capital-constrained investor in the economy did not build either; whether irreversibility independently binds cannot be settled here. Kazakhstan operates as a corridor, not a factory.

Keywords: trade reorientation, demand shocks, investment under uncertainty, irreversibility, industrial policy, value added, institutional voids, transit economies, Kazakhstan. **JEL:** F14, F15, E22, D25, O14, O24, O33, P33.

1 Introduction

When demand for a country’s tradeable output rises sharply and persistently, the textbook expectation is that supply follows: firms expand, new entrants appear, and, if the country was importing the good, some of the demand is met domestically. This logic underlies a large part of development economics, from infant-industry and import-substitution arguments to modern work on global-value-chain entry and the investment effects of trade shocks. Its premise is that a demand signal, once clear enough and durable enough, is acted on by investment.

After February 2022 Kazakhstan received an unusually clean version of such a signal. The disruption of direct trade between Russia and its principal partners reallocated a set of trade flows through third countries. Chupilkin et al. [2026] document the pattern: a sharp fall in EU exports to Russia alongside a matching rise in EU exports to Armenia, Kazakhstan and the Kyrgyz Republic, with product-level and mirror-statistics evidence of a corresponding rise in those countries’ exports to Russia; Chupilkin et al. [2025] quantify the substitution. A policy literature documents the same reallocation [Simola, 2023, Bilousova et al., 2024, Wiśniewska, 2025]. For Kazakhstan the reorientation is concentrated in a well-defined set of technology-intensive products and is large: at the monthly level, exports to Russia in these lines rose roughly tenfold ($3.5\times$ on a selection-free, externally compiled priority list not chosen on this outcome; Section 5). The routes, the demand and the products are all observable in the trade data. We treat this as a case study (a large, sector-specific, well-identified demand shock) and ask whether it induced the domestic investment response the textbook predicts.

We proceed in three steps.

First, we measure the shock. Using UN Comtrade at HS6, annual for 2018–2025 and monthly for 2019–2025, we identify from the data a “surge basket” of 29 product lines in which the outflow to Russia and the inbound flow (Western reporters plus China) both at least doubled from 2019–2021 to 2022–2024. The outbound aggregate breaks sharply in 2022m5 (monthly sup- F of 561, or 326 under a Newey–West correction, with an unchanged confidence interval); the inbound flow breaks in the same month but more weakly, and not at all in the noisier annual series, where a large pre-existing China flow dominates. Armenia and Türkiye show the same 2022 break, and the Kyrgyz Republic and Georgia show elevated exports from 2022 on a noisier series, all with no comparable domestic reform (Appendix E). A difference-in-differences against control lines gives $\gamma = 2.44$ on the data-driven basket, but a selection-rule-matched permutation places the estimate inside its own null distribution, so we do not read that coefficient as an identified magnitude. The externally compiled priority

list breaks at the same date, and offers partial corroboration at a smaller magnitude: its own level rise is a third as large ($3.5\times$ against the surge basket’s $11.6\times$), and the 26 priority-list lines that are not already in the surge basket – estimated against a control purged of the surge basket itself, rather than the full panel – show a smaller but statistically distinguishable rise on the outbound flow to Russia ($\gamma = 1.94$, wild-cluster bootstrap $p = 0.035$; own level rise $2.4\times$; pre-trend joint Wald $F = 0.75$, $p = 0.528$), though not on the inbound flow ($\gamma = 1.12$, $p = 0.238$). The priority list’s larger, shared-line coefficient is not an artifact of the 24 lines it shares with the surge basket alone, but its magnitude is not independent of the surge basket either: the residual lines move in the same direction at a smaller magnitude. A placebo on the largest civilian lines runs the other way, with no overlap between the lines it fake-treats and the surge basket.

Second, we measure what Kazakhstan retains. On the incremental flow to Russia (about \$479m over 2022–2025), the unit-value and weight evidence does not sharply distinguish the re-export flow from ordinary Kazakh trade. Propagating a wholesale-and-freight margin (6–14%, a band we choose rather than estimate) through the input–output table (OECD ICIO) implies domestic value added of about **5–11 cents** per dollar rerouted, against roughly three-quarters for a dollar of domestic manufacturing output. Because the input–output step does almost no work, the headline scales one-for-one with the margin: it stays below one-fifth of a produced dollar for any margin below a fifth (Section 6). Customs duty on the incremental flow is well under 1% of customs revenue, and the gross reoriented flow is 0.2–0.3% of GDP a year.

Third, we ask whether the shock induced investment. We find no response the data can detect. We assemble deal-level data on mergers, acquisitions, and private-equity and venture transactions in Kazakhstan for 2015–2025 from three commercial databases (Capital IQ, PitchBook, Preqin), together with the aggregate investment activity of Kazakhstan’s state investment corporation as reported in Qazaqstan Investment Corporation et al. [2026]. Deal-making in manufacturing of tradeable goods, transport and logistics, and distribution runs at essentially its pre-2022 rate, if anything lower after 2022; with four post-2022 years the test rules out a large response, not a modest step-up, and the null is in any case overdetermined: the January 2022 unrest, the cross-border compliance and reputational exposure that transactions of this kind carry, and the nationalisation wave would each depress entry. Reported deal value rises, but is concentrated in ownership transfers of existing or distressed assets (two of the largest are nationalisations), while greenfield and new-capacity transactions show no post-2022 increase.

We interpret the non-response with a framework in which the supply response is the product of three gates: *irreversibility* (does anyone build for a shock this uncertain and

policy-contingent, given the option to serve the demand through trade?), *market access* (is domestic transformation required to reach the final market?), and *institutions* (can the projects that clear the first two be financed?). For Kazakhstan the market-access gate is plainly open (a customs union with the destination makes trade a close substitute for production), which is close to sufficient for a null only jointly with a small transformation margin, not on its own (Section 3), and which also means Kazakhstan forgoes the technology transfer a rules-of-origin regime would have forced [de Souza et al., 2024]. The case is also inconsistent with capital-market institutions as the binding constraint for this particular null: the state investment corporation, which faces neither an external-finance nor an exit constraint, did not build for the reallocation either. Whether the irreversibility gate *independently* binds we cannot settle here: a within-country comparison (an assembly response to the durable passenger-vehicle demand shock, none to the re-export shock) is suggestive but confounded, and because the re-export flow in fact persisted through 2025, a 2022 entrant faced uncertainty about its duration rather than an observed short shock. The institutional gate does explain a separate fact: why the adjacent durable opportunities the shock reveals go unfinanced by private capital.

Contribution. The paper makes three contributions. (i) Its primary contribution is empirical: to our knowledge it is the first host-economy analysis of what a transit economy retains from a post-2022 trade reallocation and of whether it induced investment, extending a literature that establishes the fact and scale of the reallocation to what the intermediary retains from it, and to 2025. It also connects to Bown and Crowley [2007]’s finding that import-restricting trade policy deflects a foreign country’s exports into third markets: the present case takes that deflected flow as its starting point and asks what the receiving economy does with it. It shows that a large, sector-specific tradeable-demand shock produced no detectable investment response, a fact that speaks to when trade shocks induce a domestic supply response and when they do not [Juhász, 2018, Bloom et al., 2016, Juhász et al., 2024, Gopinath and Neiman, 2014]. (ii) It organises the two standard readings of a missing supply response (investment under uncertainty and irreversibility [Dixit and Pindyck, 1994, Handley and Limão, 2015, 2017], and institutional voids in the capital market [Khanna and Palepu, 1997, 2000, Rajan and Zingales, 1998]) into a single decomposition, and adds a customs-union market-access gate under which trade substitutes for production, engaging the alternative that an export-platform FDI response should instead predict [Brainard, 1997, Helpman et al., 2004, Ekholm et al., 2007]. As Section 8 sets out, the case establishes an open market-access gate and finds evidence inconsistent with the institutional gate as the binding constraint for this null, and leaves the irreversibility gate undetermined; the

decomposition is an organising device, and we do not estimate it. (iii) Methodologically, it combines public trade data, an input–output table and multi-source commercial deal data into a replicable measure of what a transit economy captures, cross-checking the deal counts across databases.

The rest of the paper is organised as follows. Section 2 describes the setting; Section 3 sets out the framework; Section 4 the data, including the deal-database construction; Section 5 measures the reorientation; Section 6 what Kazakhstan retains; Section 7 the (non-)investment response; Section 8 asks which gate binds; Section 9 turns to the moderators and external validity; Section 10 identifies where investment would raise value capture; Section 11 draws the policy implications; and Section 12 concludes.

2 Setting

Kazakhstan is a founding member of the Eurasian Economic Union and, before it, the 2010 customs union with Russia and Belarus, which removes the customs frontier between Kazakhstan and Russia: goods that clear customs on entry to Kazakhstan move onward to Russia without a further border, and a supply from Kazakhstan to Russia is treated as intra-union. The union aligned Kazakhstan’s external tariff schedule with Russia’s, largely at the Russian duty levels prevailing at the time, and measurably shifted the sourcing of its imports away from China and toward within-union partners, though the estimated magnitudes are modest [Isakova et al., 2016]. Kazakhstan is also the land bridge between China and Russia and, via the Trans-Caspian corridor, between China and Europe, though as a landlocked economy its overland logistics costs are high and, more binding still, unpredictable [Arvis et al., 2010]. These features make it a natural transit route for goods re-exported without transformation: because it shares a customs union with the destination, serving the final market from Kazakhstan requires no domestic processing.

The land route has also become more valuable for reasons unconnected to any single trading partner. Since 2022 the northern land corridor through Russia has been closed to most Western supply chains, and over 2023–2025 the principal maritime routes were disrupted in turn: draft restrictions on the Panama Canal, re-routing around the Red Sea and periodic risk in the Strait of Hormuz. Each raises the value of an overland China–Europe alternative, and the shortest route that avoids both Russia and the maritime chokepoints runs through Kazakhstan and across the Caspian. We return to what this implies for investment policy in Section 11.

Several things other than the reorientation changed in Kazakhstan around 2022, and each is a potential confound. The first is the “New Kazakhstan” reform programme associated

with the presidency of Kassym-Jomart Tokayev, an outward-oriented agenda that gathered pace in 2022 with a June constitutional referendum and a November snap election. A reform of that kind would raise trade and investment broadly and gradually. What we observe is the opposite: a response concentrated in a narrow set of products, discontinuous in mid-2022, with a civilian control basket that moves far less, and the same disruption evident in Armenia and Türkiye by 2022 (the Kyrgyz Republic and Georgia elevated on noisier series; Appendix E) under no comparable reform.

Three further shocks bear specifically on the *investment* null and cannot be differenced away: the January 2022 unrest and state of emergency, in the first month of the post-period; the cross-border compliance and reputational exposure that transactions of this kind carry, which deters exactly the foreign acquirers who might have answered the shock; and the 2022–2025 wave of nationalisations, which both crowds the state investment pipeline and signals ownership risk to private entrants. We return to these where they matter, in Sections 7 and 8. They make the investment null *over*-determined, though this also means we cannot cleanly attribute it to the trade shock alone.

3 A framework: when a demand shock builds capacity

Consider a potential entrant deciding whether to build capacity to serve a newly enlarged demand for a tradeable good that the country currently imports. The good can be re-exported to the final market without domestic transformation, which is the relevant case for a member of a customs union with the destination. The entrant has two ways to serve the demand. It can *trade*: buy the good abroad and resell it into the final market, earning a margin μ_T per unit, with no sunk cost and the ability to stop costlessly if demand falls. Or it can *produce*: sink an irreversible cost I to build capacity, then earn a margin μ_P per unit on domestic output for as long as it operates, with the capacity worth only scrap if the demand disappears.

Let the enlarged demand support quantity Q per period and persist each period with probability ρ (so expected duration $1/(1 - \rho)$), and let r be the discount rate. Trading is always weakly profitable while the demand lasts, with present value $V_T = \frac{\mu_T Q}{1 - \rho/(1 + r)}$. Building is profitable only if

$$\underbrace{\frac{\mu_P Q}{1 - \rho/(1 + r)}}_{\text{PV of production margin}} - I > \underbrace{\Omega(\sigma, \rho)}_{\text{option value of waiting}} + V_T, \quad (1)$$

where Ω is the value of the option to defer, increasing in the uncertainty σ around the shock's

size and duration and decreasing in ρ [Dixit and Pindyck, 1994]. The policy-contingent uncertainty that raises Ω here is the same channel Handley and Limão [2015, 2017] formalise and estimate for trade-policy uncertainty depressing export entry and investment; (1) adds the re-export outside option V_T to that channel, which is the addition the market-access gate below turns on. I and Ω are the terms this literature owns: sunk market-entry costs generate the same hysteresis in export participation [Roberts and Tybout, 1997], and their structural estimates put entry costs an order of magnitude above per-period profits and highly heterogeneous across producers, which is why credible persistence, not merely large expected demand, is what moves the entry margin [Das et al., 2007]. Four observations follow.

(i) *The irreversibility gate.* The right-hand side of (1) is large exactly when the shock is transitory (low ρ), uncertain (high σ , e.g. because it is policy-contingent), and the technology is sunk-cost-intensive (high I). When any of these is extreme, no entrant builds and the investment response R is near zero, regardless of the other conditions. This is a statement about *whether anyone builds*, and it does not depend on who the entrant is or how deep the capital market is.

(ii) *Trade is the outside option, not just “wait”.* Because the good is re-exportable, the alternative to building is not idleness but V_T : the entrant can serve the demand through trade indefinitely. The demand shock raises the return to *both* margins. When the shock is transitory and μ_T is available at low cost, (1) fails and the supply response is trade rather than production. A corollary: if $\mu_P \leq \mu_T$ (a domestic plant’s per-unit transformation margin is no larger than the margin a re-exporter already earns), building is dominated before Ω is added. We do not identify μ_P or μ_T directly; two pieces of evidence are suggestive, not dispositive, that $\mu_P \lesssim \mu_T$ for this sector. First, its domestic value-added multiplier (0.69; Section 10) is a share of gross output accruing domestically, not a per-unit margin, so it bears on μ_P only loosely: a low multiplier is consistent with a low transformation margin but does not measure it. Second, the sector’s near-zero domestic base ($\approx$ \$230m of electronics output) is itself the outcome the paper is trying to explain, so citing it as evidence for $\mu_P \lesssim \mu_T$ risks circularity — an absent industry is equally what a customs union with no comparative advantage in the sector would produce, independent of any margin comparison. We return to this alternative reading, and what would distinguish it, in observation (iii).

(iii) *The market-access gate.* If serving the final market instead *required* domestic transformation (binding rules of origin, local-content rules, or simply a customs border that a pure re-export cannot cross), then $V_T \rightarrow 0$ and (1) is far more likely to hold. Membership of a customs union with the destination opens this gate and is what makes trade a viable substitute for production.

This is not the only prediction a customs union makes, however, and the case for the gate having bound here is joint, not stand-alone. The export-platform FDI literature [Brainard, 1997, Helpman et al., 2004, Ekholm et al., 2007] shows that lower trade costs with a destination can pull *production* toward the member that serves it tariff-free just as readily as they enable pure re-export: a customs union lowers the cost of shipping a finished good onward from a local plant exactly as it lowers the cost of re-export, so μ_P and V_T can rise together, and (1) on its own does not say which effect dominates. An open market-access gate drives R toward zero only jointly with a small transformation margin ($\mu_P \lesssim \mu_T$); membership in a customs union alone does not deliver that combination, which is why observation (ii) treats the margin comparison as suggestive rather than settled. What would distinguish the two readings is direct evidence on μ_P : the transformation margin actually available to a Kazakh producer in a customs-union-facilitated line elsewhere in the economy, or a comparison with an economy at a similar income level that lacks a customs union with its main trading partner. We do not have that comparison. The case here identifies an open market-access gate, but it cannot rule out that Kazakhstan’s absence of an electronics industry is equally consistent with the country simply lacking a comparative advantage in the sector, customs union or not.

The same logic runs through the imported-input intensity of the would-be domestic producer: where domestic production would itself lean heavily on imported components (the roundabout structure of Gopinath and Neiman [2014]), the value a local plant adds over a pure re-export is smaller still, and the transformation margin $\mu_P - \mu_T$ narrower. The gate also governs *dynamic* gains. de Souza et al. [2024] show, using Brazilian firm-to-firm technology-transfer records, that a foreign supplier facing a lower tariff shifts from *transferring* technology into the market toward *exporting* goods to it, and that learning from imported goods is less efficient than learning from a transfer, so import liberalisation raises static welfare but slows diffusion and productivity growth. An open market-access gate is the limiting case: with a pure re-export, the foreign firm has no reason to transfer anything, and the host forgoes both the static value added of Section 6 and the learning that would come with a plant.

(iv) *The institutional gate.* Suppose (1) *does* hold for some project, a durable opportunity with modest sunk costs. It is still realised only if the entrant can finance I and, for an outside equity investor, exit. Where the capital market lacks growth equity and a functioning exit, projects that clear the irreversibility gate are not funded; the capital that does move is captive (retained earnings, or a state or development-finance investor) and is allocated by criteria other than R [Khanna and Palepu, 1997, 2000, Rajan and Zingales, 1998].

The three conditions can be summarised as a product, though only two of the four obser-

vations above identify a factor that alone drives the investment response to zero regardless of the others: an extreme irreversibility configuration, or a capital market that cannot fund the projects that pass. The market-access gate is different, as observation (iii) argues: an open gate (transformation not required, so V_T is large) drives R toward zero only jointly with a small transformation margin, not on its own. Writing

$$R = \underbrace{f(\rho, \sigma^{-1}, I^{-1})}_{\text{irreversibility}} \times \underbrace{h(\text{transformation required})}_{\text{market access}} \times \underbrace{g(\text{financial depth, exit, private/state mix})}_{\text{institutions}} \quad (2)$$

summarises the case’s argument; it is not a result derived from (1) (whose threshold is on a sum, with the three considerations interacting inside a single indicator rather than entering as separable multiplicative terms). We do not claim, and do not need, that the three conditions are separately *necessary*: each of the arguments made below — irreversibility, market access taken jointly with the transformation-margin evidence, and institutions — is argued for Kazakhstan on its own terms in Sections 5–8, rather than derived from (2). Those sections show that for Kazakhstan the market-access gate is open and, jointly with the sector’s low transformation margin, close to sufficient for $R \approx 0$ (subject to the alternative reading noted above); that the state investment corporation, which faces neither an external-finance nor an exit constraint, did not build for the reallocation either, which is inconsistent with capital-market institutions as the binding constraint for this null; and that whether the irreversibility gate independently binds cannot be settled. Section 9 and Section 10 show that the institutional gate does explain a separate fact: the durable adjacent opportunities the shock reveals go unfinanced.

A mediation restatement. The same structure can be put in mediation terms; we do so as an interpretive device, not as a model we estimate, since the data below have no firm-level panel linking the trade response to the investment response. Let the demand shock X bear on the investment response Y through a direct path $X \rightarrow Y$ and an indirect path $X \rightarrow M \rightarrow Y$, where the mediator M is the volume of re-export activity the shock generates. The shock raises M ; higher M substitutes for domestic capacity, so it works against Y . When the market-access gate is open ($h = 1$, no transformation required), trade is a close substitute and nearly all of the shock is carried by M : the channel that would raise Y is barely opened, and the net effect on investment is close to nil. When the gate is closed ($h < 1$), M absorbs less of the shock and part of X bears on Y directly, with that residual governed by the irreversibility parameters (ρ, σ, I): whether the un-absorbed demand is large and durable

enough to clear (1). Kazakhstan is the polar case: $h = 1$, and the unit-value evidence of Section 6, though it does not identify the margin, is consistent with a re-export price no higher than the delivered cost of the input, so little is left for the direct path even before the irreversibility gate is applied.

4 Data

Trade. UN Comtrade, HS 6-digit. Annual 2018–2025 for magnitudes; monthly (authenticated API) for break timing and the event study. Kazakhstan suspended monthly reporting to Comtrade after February 2024 and resumed in January 2025, so the monthly series runs 2019m1–2024m2 and 2025; the 2024 gap is covered by the annual data and by mirror flows. We use Kazakhstan’s reported exports to Russia (complete) and, for the inbound side, a partner-reported (mirror) measure: the sum of exports to Kazakhstan reported by the EU-27, the United Kingdom, the United States, Japan, Korea, Switzerland and Norway (“Western”) *and China*, because Kazakhstan’s own import reports are incomplete for 2020–2022. China supplies about two-thirds of this inbound value and its flow to Kazakhstan was large and rising well before 2022; we therefore also track the Western component on its own, since that is the part that moves with the reorientation. Mirror data overstate Kazakh absorption for goods declared for Kazakhstan and then moved onward (the onward-movement bias that Chupilkin et al. [2026] use to identify the reorientation in the first place), a bias we return to in Sections 5.3 and 6. Neighbour comparisons (Armenia, the Kyrgyz Republic) and the economy-wide imports-by-chapter series used in Section 10 are from the same source.

Product sets. We take 75 HS6 lines: 50 codes from an externally compiled list of technology-intensive HS6 items published in February 2024 (the “priority list”) plus 25 clearly civilian control codes. We define the **surge basket** from the data as the 29 lines in which the mean inbound flow (West + China) and the mean outflow to Russia each at least doubled from 2019–2021 to 2022–2024, and exceeded \$0.2m (inbound) and \$0.1m (outbound) in the post period, computing the ratio with \$10,000 added to numerator and denominator so that a line rising from near-zero does not qualify on a rounding artefact. Twenty-four of the 29 lines are on the priority list; the codes, tiers and control set are in Appendix B (Table B.1). Because the basket is selected on the post-period outcome, we also run every specification on the priority list, which is defined on product characteristics rather than on Kazakh flows (though, compiled in 2024, it is not independent of the phenomenon it helps describe), and we benchmark the difference-in-differences against a selection-rule-matched permutation test (Section 5.3).

Input–output. OECD Inter-Country Input–Output tables, 2023 edition, Kazakhstan block, 45 ISIC rev. 4 industries, 2019. We form the domestic technical-coefficient matrix A_d from the Kazakhstan-on-Kazakhstan intermediate block, the domestic Leontief inverse $L_d = (I - A_d)^{-1}$, and the direct domestic value-added coefficients v_d ; the domestic value-added multiplier for sector j is $(v'_d L_d)_j$. As a robustness check we repeat the calculation on the Kazakhstan Bureau of National Statistics symmetric input–output table (up to 68 products, 2023 reference year).

Deal data. Deal-level records of mergers, acquisitions, private-equity buyouts and growth transactions, and venture rounds involving Kazakhstan-domiciled targets, 2015–2025. The extract combines Capital IQ, PitchBook and Preqin ($N \approx 500$ after de-duplication), each record carrying its native source deal identifier, classified to sectors and flagged for greenfield versus ownership transfer. The exact query specification for the three databases (geography filter, deal types, date range, industry taxonomy, extraction date) is in Appendix A, and Table 3 reports each one’s coverage. All are commercial but widely available under academic licence, and the replication package lists the deal identifiers, so the extract is reconstructible. QIC investment activity is taken from Qazaqstan Investment Corporation et al. [2026], which reports it in aggregate (about 98 investment records, roughly \$2.2bn, 2015–2025), by sector and year, and names a handful of individual investments in its case studies; no project-level QIC register is public, and the paper uses only the report’s aggregate, sector-timing and named-investment evidence.

Macro. World Bank World Development Indicators.

5 The demand shock

5.1 Magnitudes

Table 1 reports the surge basket. Kazakhstan’s exports to Russia in these lines were \$7–17m per year in 2018–2021; they were \$128m in 2022 and \$119–145m in 2023–2025, a rise of roughly an order of magnitude. The inbound flow (West + China) ran near \$440m per year before 2022 and roughly \$0.8–2.4bn after, but that series is dominated by China, whose exports to Kazakhstan were already large and rising before the war (and are only partly reported for 2024–2025). The *Western* component of the inbound flow (the part aligned with the reorientation) rose from about \$170m per year to \$340–440m, a factor of about 2.3.

Table 1: Surge-basket trade, USD million per year. Inbound is a partner-reported (mirror) measure. China’s 2024 figure is large and preliminary; its 2025 figure is not yet reported, so the 2025 West + China total is the Western component only. The priority-list row is the externally compiled 50-HS6 list (not selected on this outcome), shown as a selection-free comparator to the surge basket’s "roughly tenfold" rise.

	2018	2019	2020	2021	2022	2023	2024	2025 [†]
West + China → KZ (mirror)	394	427	483	470	837	1,373	2,360	363 [†]
<i>of which Western</i>	162	162	168	184	337	444	419	363
Kazakhstan → Russia	17	12	7	8	128	145	119	133
<i>Priority list, overlapping (50 HS6)</i>	82	89	59	61	306	293	229	187

[†] 2025 West + China is the Western component only; China has not yet reported. Priority-list pre-2022 mean \$72.7m, post-2022 mean \$253.7m, ratio 3.5×, against the surge basket’s 11.6× (Section 5).

5.2 Timing and the event study

Monthly, the outbound series (Kazakhstan’s exports to Russia) breaks sharply in 2022m5 (Bai–Perron [Bai and Perron, 1998], 95% CI 2022m4–2022m6; sup- F [Andrews, 1993] of 561 under a homoskedastic covariance, against critical values in the low tens), following the March-2022 sanctions with a short lag and preceding every political-calendar date. Because the series is persistent, a homoskedastic covariance can overstate the sup- F and understate the confidence interval; under a Newey–West [Newey and West, 1987] correction the sup- F falls to 326 but the 95% CI is unchanged (2022m4–2022m6), so the timing argument does not depend on the choice of covariance estimator. The inbound series (West + China) breaks in the same month (95% CI 2022m4–2022m6, HAC sup- F 15.5 against 24.1 homoskedastic, CI unchanged), with a second, smaller break in late 2023 as trade-monitoring measures tightened, and the Western component alone breaks in 2022m5 more sharply (95% CI widens by one month to 2022m3–2022m6 under the HAC correction). On the *annual* panel the outbound and Western-inbound breaks are again dated to 2022. The West + China inbound break is not significant, because the large pre-existing China flow swamps the 2022 movement at annual frequency; the reorientation is therefore best seen in the outbound series and the Western inbound.

The sup- F is computed without a HAC correction and the series is highly persistent, so we read it as evidence of a break and its *date*, not as calibrated inference on the break’s size. The event study and the difference-in-differences carry the quantitative weight. Figure 1 plots the monthly series. Figure 2 is a difference-in-differences event study at the HS6×year level (surge basket vs. control lines, reference year 2021, 95% confidence intervals clustered by HS6). The pre-treatment coefficients are close to zero and jointly insignificant ($p = 0.78$ outbound, $p = 0.42$ inbound). The outbound coefficient jumps in 2022 and stays elevated

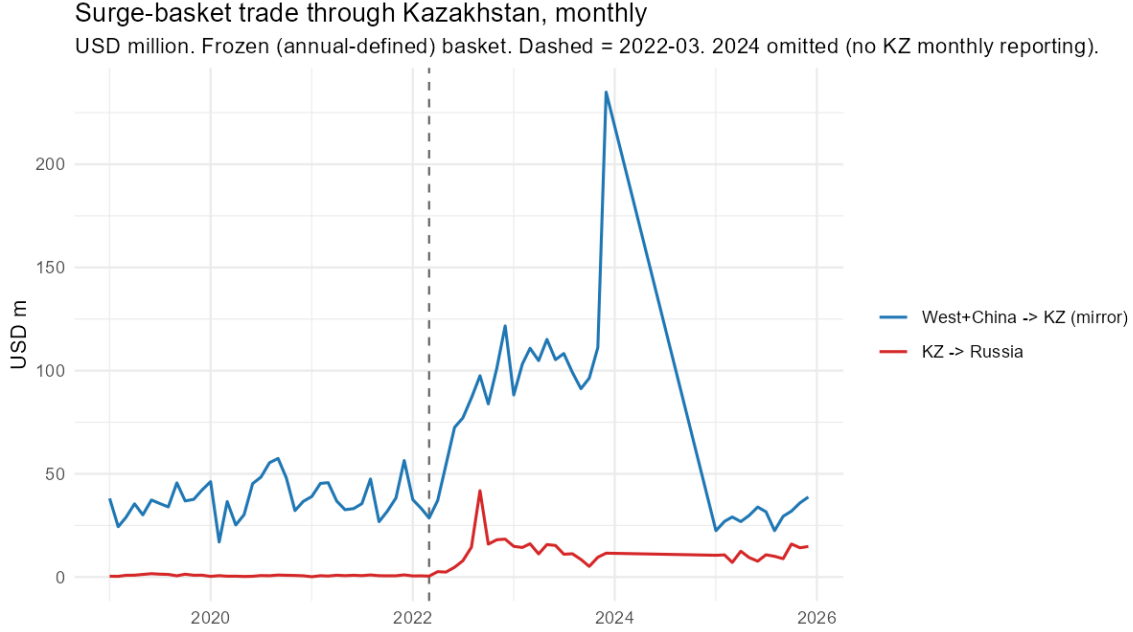


Figure 1: Surge-basket trade through Kazakhstan, monthly. USD million; blue = West + China mirrored exports to Kazakhstan, red = Kazakhstan’s exports to Russia. 2024 omitted (Kazakhstan’s monthly reporting to Comtrade is suspended after 2024m2). Dashed line: March 2022. Source: UN Comtrade.

through 2025 at about +2.4 asinh points (the Poisson level estimate is larger and is reported in Table 2; we do not read it as a headline magnitude, for the reasons in Section 5.3), and the inbound coefficient follows a similar path with wider intervals. That the break precedes the June referendum, that the response is confined to a narrow product set, and that Armenia and Türkiye show the same 2022 break, with the Kyrgyz Republic and Georgia also elevated from 2022 on noisier series, and none with a comparable domestic reform (Appendix E) together separate the reorientation from the “New Kazakhstan” agenda.

5.3 Difference-in-differences

Specification and identifying assumptions. On the annual panel we estimate $\text{asinh}(y_{it}) = \gamma (\text{treated}_i \times \text{post}_t) + \mu_i + \lambda_t + \varepsilon_{it}$ with HS6 and year fixed effects and standard errors clustered by HS6 (75 lines, 8 years; reference year 2021). Zeros are common on the outbound leg before the shock and rare after (23% of surge-basket cells are zero in 2018–2021 against 3% in 2022–2025), so we report the asinh specification alongside a Poisson (PPML) estimate in levels [Santos Silva and Tenreyro, 2006], which handles the zeros without a transformation and weights by size. y is measured in raw USD, so with a substantial zero-and-near-zero pre-period mass the asinh coefficient does not have a single clean percentage-change reading

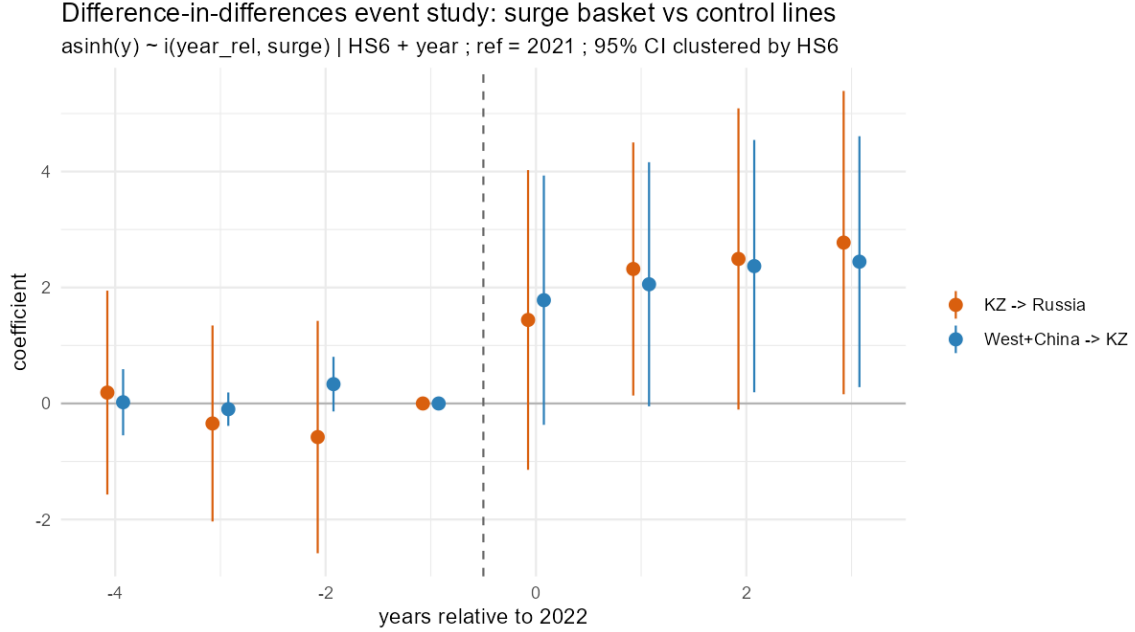


Figure 2: Difference-in-differences event study, surge basket vs. control lines: $\text{asinh}(y_{it}) = \sum_{k \neq -1} \beta_k (\text{surge}_i \times \mathbf{1}[t - 2022 = k]) + \mu_i + \lambda_t + \varepsilon_{it}$, HS6 and year fixed effects, 95% confidence intervals clustered by HS6, reference year 2021. Source: UN Comtrade.

across cells [Bellemare and Wichman, 2020]; the two specifications agree on sign and significance but not on implied magnitude (the asinh point estimate corresponds to a level effect several times larger than PPML’s $3.5\times$), consistent with the general warning against reading log-like transformations of a variable with a nontrivial zero mass as percentage effects [Chen and Roth, 2024], which is a reason we lean on the sign and the structural breaks rather than on the asinh coefficient’s size (Table 2). Identification requires parallel paths absent the shock (a joint test of the three pre-period interaction coefficients does not reject: $p = 0.42$ inbound, $p = 0.78$ outbound) and no anticipation (dropping 2022 entirely, or setting the treatment year to 2023, moves γ only modestly, to 2.71; see below). The civilian control lines are not perfectly clean: their exports to Russia also break in 2022 (see “The reform confound” below), so γ is if anything attenuated relative to the economy-wide effect. The surge basket is *selected on the post-period outcome*, which inflates γ mechanically; the selection-rule-matched permutation test below, not the cross-sectional regression, is what speaks to this and to the cross-sectional dependence a single common treatment date induces — exactly the small-number-of-policy-changes setting Conley and Taber [2011] and Ferman and Pinto [2019] analyse, and our randomisation-inference approach follows their prescription of resampling treatment assignment rather than trusting asymptotic cluster-robust inference in this regime. We also report wild-cluster-bootstrap p -values [Cameron et al., 2008] (few clusters, within-HS6 serial correlation) and Holm-adjust across the four surge-basket outcomes.

Both the analytic and the bootstrap standard errors cluster within HS6 but treat lines as independent; with a single common treatment date and a shock that hits related product lines together, that is optimistic. The randomisation-inference p -values below do not rely on it (they resample whole treatment histories) and are the inference we lean on.

Results. Table 2 reports γ for the surge basket, the externally compiled priority list, and a placebo. On the surge basket, exports to Russia rise by $\gamma = 2.44$ ($p = 0.013$; wild-bootstrap $p = 0.010$; Holm-adjusted $p = 0.039$; Poisson in levels $3.5\times$, $p = 0.02$). The inbound flow (West + China) rises by $\gamma = 2.10$ ($p = 0.05$, not significant after Holm adjustment); the Western component alone by 1.73 ($p = 0.08$); Kazakhstan’s own reported imports by 1.88 ($p = 0.004$, Holm $p = 0.016$). The externally compiled priority list, which is not selected on the Kazakh outcome, gives the same pattern ($\gamma = 2.88$ outbound, 2.12 inbound). The outbound effect survives two further checks: adding size-decile $\times$ year fixed effects (pre-2022 inbound size) leaves $\gamma = 2.29$ ($p = 0.003$), and dropping 2022 raises it to 2.71 ($p = 0.012$). The same size $\times$ year controls cut the West + China inbound coefficient to 1.28 ($p = 0.05$), consistent with the inbound leg being the weaker of the two throughout. Tightening the selection threshold from a doubling to $2.5\times$ or $3\times$ raises the outbound γ to about 3.1. Loosening it to $1.5\times$ (40 lines) lowers it to 1.6 ($p = 0.08$). A leave-one-HS2-out jackknife shows the effect is carried by HS 85: dropping it (13 of the 29 lines) leaves $\gamma = 1.36$ ($p = 0.21$); the other six chapters, dropped one at a time, give $\gamma = 2.51$ (drop HS 39), 2.48 (HS 40), 2.67 (HS 61), 2.28 ($p = 0.06$; HS 84), 3.16 (HS 90) and 2.59 (HS 96), all significant at 5% except HS 84 at 6%. The point estimate thus depends materially on the electrical-machinery lines, which is where the reorientation is concentrated; the sign and the sub-5% significance do not, except when the largest chapter is removed.

Three qualifications keep us from resting identification on this cross-sectional regression (the full battery is in Appendix C). First, the selection-rule-matched permutation: when we impose the null by permuting the year labels within each HS6 and re-applying the doubling rule, the rule selects on average five lines (never as many as the 29 observed, so the *existence* of the surge basket is not a chance artefact, $p < 0.001$); a trend-preserving version that only shifts the break off calendar-2022, keeping each line’s autocorrelation and level trend, gives the same result (null mean five lines, $p < 0.001$), so it is the 2022 alignment and not a pre-existing trend that the selection picks up. But the observed $\gamma = 2.44$ sits squarely inside the rule-matched null distribution of the coefficient itself (null mean 2.79, s.d. 1.50; $p = 0.58$ for $|\gamma^{\text{perm}}| \geq |\gamma^{\text{obs}}|$), so the coefficient’s *magnitude* is not separable from the selection and we do not read it as an identified effect size. Second, the two selection thresholds bind differently: 55 of the 75 candidate lines clear the outbound doubling but only 41 clear the inbound one,

so requiring both is a real screen on the import side, not just on the outflow to Russia; this is a partial check against selecting the basket purely on the outbound outcome that the difference-in-differences then uses. Third, the placebo row: assigning fake treatment to the largest civilian lines gives $\gamma = -0.95$ ($p = 0.001$; the large consumer-goods lines contracted after 2022 as the tenge depreciation would predict), which shows the control pool is not homogeneous in trend. The surge coefficient’s survival of the size-decile $\times$ year controls addresses this, and we do not read the placebo as corroboration. Taken together, the cross-sectional difference-in-differences documents the outbound reorientation but does not identify it on its own; the structural breaks (Section 5) and the neighbour comparison below carry that weight. The mirror-gap outcome (partner-reported inflow minus Kazakh-reported imports) does not move ($\gamma = 0.54$, $p = 0.81$).

The reform confound. Three things separate the reorientation from the outward-oriented policy agenda of 2022. First, the civilian control basket behaves in a way a broad reform cannot explain: on the annual data its inbound flow shows no 2022 movement at all, and while its exports to Russia do shift (a general Russia-trade effect), the shift is far smaller than the surge basket’s on the same test, and the monthly surge-basket outbound break is much sharper still. Second, the outbound break is dated 2022m5, before the June referendum and the November election. Third, and most directly: Armenia’s surge-basket exports to Russia rose from \$7–15m per year to \$71m in 2022 and \$93m in 2023, and the Kyrgyz Republic’s from \$1–7m to \$12m and \$30m, on the same timing, neither country having a “New Kazakhstan” reform. The direction is also the right one: the goods come in from Western reporters and move onward to Russia, not the reverse.

6 What Kazakhstan retains

6.1 The retained trade margin

We measure what Kazakhstan retains on the *incremental* flow (exports to Russia above the pre-2022 baseline, about \$479m cumulatively over 2022–2025), since that is the part the reorientation drove. The \$479m increment is measured against a flat counterfactual at the 2018–2021 mean (\$11m per year); because the pre-2022 outbound series is itself declining (\$17m $\rightarrow$ \$8m), that is the conservative choice: extrapolating the pre-trend would put the counterfactual near zero by 2024 and raise the increment to about \$521m, widening every downstream figure by roughly a tenth. Set against the incremental *Western* inbound of about \$887m, the flow-through is roughly one-half; set against the incremental West + China

inbound of about \$3.2bn, it is one-sixth.¹ The inbound leg is the more weakly identified of the two (its DiD coefficient loses significance under size×year controls and after multiple-testing adjustment, Section 5.3), so the flow-through ratio should be read as indicative, not precise. The remaining $\$887\text{m} - \$479\text{m} = \$408\text{m}$ of the Western increment decomposes into only about \$41m of onward exports to destinations other than Russia (Kazakhstan’s surge-basket exports to the rest of the world barely move) and about \$367m of domestic absorption, inventory and the c.i.f./mirror measurement gap, so the flow-through gap is not hidden re-export through third countries. The Western basis is the relevant one for the reorientation, and we use it here.

The unit-value evidence is consistent with a corridor and does not identify a retained margin on its own. On a c.i.f./f.o.b. basis (Kazakhstan’s f.o.b. re-export price against the c.i.f. price it records paying on import, which includes freight into a landlocked economy), the median ratio across matched HS6×period cells is **0.73** (annual, $n = 112$; 0.70 monthly). On a like-for-like f.o.b. basis, against the exporter-reported price at the partner’s border, the median is **1.6** (annual; 1.5 monthly), the difference being roughly the inbound freight. Neither ratio is distinctive (the civilian control basket gives 0.59 and 0.94 on the two bases), and the pass-through regression is weak on both (slopes of 0.20 and 0.48, significant only in the annual f.o.b. specification).

Two features cut against domestic transformation, though neither is decisive. The first is the weight ratio (re-exported tonnage over imported tonnage in the matched cell), which is 0.08 at the median and 0.11 in aggregate, close to the matched-cell value flow-through (median 0.15, aggregate 0.11). The aggregate weight ratio and the aggregate value flow-through are the same construction on different units, and here they coincide at 0.11, so a sub-one weight ratio is largely a flow-through fact, given that only part of the imported tonnage is re-exported to Russia at all, and is not evidence on weight gain per transiting unit. Eleven per cent of cells do carry a weight ratio above 1.05, on a wide interquartile range (0.02 to 0.49); under the flow-through reading that upper tail is timing and inventory rather than processing. If we restrict to the six near-pure-transit HS6×period cells where value flow-through is near one, the median weight ratio is 0.74 (0.26 and 0.94 at the quartiles, on $n = 6$), showing no sign of local input added on the goods that do transit, though on a thin slice of the basket. The Comtrade extract as pulled carries value and net weight but

¹Three flow-through ratios appear in this section, on different samples and bases: the *incremental* ratio against the Western inbound (≈ 0.54 , on aggregate 2022–2025 minus baseline) and against West + China (≈ 0.15 , “one-sixth”); the latter is computed on 2022–2024 only, since the 2025 West + China figure in Table 1 is Western-only (China not yet reported) and mixing it into a West + China increment would understate it by construction; the 2022–2025 mixed-basis figure gives the same ratio to two decimal places. The *matched-cell* aggregate ratio (≈ 0.11 , sum of re-export value over sum of inbound value across the 112 matched HS6×period cells) used in the weight-ratio discussion below is a third, separate construction.

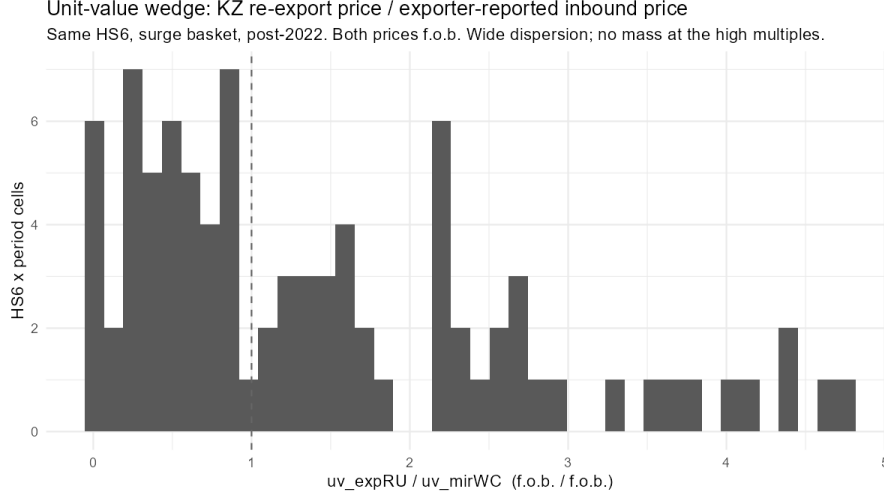


Figure 3: Unit-value wedge: Kazakhstan re-export price over the exporter-reported inbound price (f.o.b. basis), same HS6, surge basket, post-2022. Wide dispersion, with the high multiples falling in a subset of lines (tier medians near four in Tiers 2 and 4A, around one elsewhere): the pattern a scarcity markup produces, not the even, across-the-board wedge a processing margin would.

not physical quantity, so a cleaner per-unit transformation test is not available here.

The second feature is the dispersion in the f.o.b. wedge, which is not a processing signal. The wedge is uneven across tiers,² with tier medians near four in Tiers 2 and 4A but around one in Tiers 3B and 4B and only 0.47 in Tier 1; the high multiples sit on about half the matched gross flow (Figure 3). An uneven wedge concentrated in a subset of lines is the pattern of a scarcity markup on goods that became harder to source directly, not the even, across-the-board wedge a processing margin would produce. It is the pattern Feenstra and Hanson [2004] document for Hong Kong’s entrepôt trade in Chinese goods, where the re-export markup is largest where the intermediary adds matching and scarcity value rather than physical transformation; the wedge here is if anything smaller and noisier. The value-weighted aggregate wedge is below one, dominated by a handful of high-inbound-value cells and by outbound under-invoicing [Fisman and Wei, 2004]. The wedge does not pin the retained margin, which we take from the national accounts next.

6.2 Input–output propagation

Let m be the wholesale-and-freight margin Kazakhstan retains on a transiting good as a share of its value, $\bar{v}^{TT}$ the mean domestic value-added multiplier for its trade, transport and

²The priority list is published with the goods grouped into tiers (Tier 1–2 the most sensitive, Tiers 3–4 progressively less so); we carry the published tier labels unchanged and use them only to describe where in the list the high wedges fall. Tier 1 = HS 8542 (integrated circuits); Tier 2 adds transceivers, capacitors, connectors; Tiers 3–4 are broader machinery, instruments and components.

warehousing sectors, and $\bar{v}^M$ that for manufacturing. Domestic value added per dollar of gross rerouted flow is $m \bar{v}^{TT}$; per dollar of domestic manufacturing output it is $\bar{v}^M$ [Koopman et al., 2014, Johnson and Noguera, 2012]. From the OECD ICIO Kazakhstan block, $\bar{v}^{TT} = 0.79$ and $\bar{v}^M = 0.76$, close enough that the propagation step does little work: the result is essentially m against the full domestic value $\bar{v}^M$.

We take m to be what a trading intermediary retains when it buys a good wholesale and re-sells it without a retail leg or a physical transformation: the c.i.f./f.o.b. freight and insurance wedge plus a wholesale (not retail) trade margin. The Kazakhstan Bureau of National Statistics resources table [Bureau of National Statistics, Agency for Strategic Planning and Reforms of the Republic of Kazakhstan, 2024] bounds it: the *transport* margin on machinery and electronics is about 1% of basic-price value, and the *full* trade-and-transport margin (which also carries the retail leg the corridor never performs) is about 49%. A wholesale-and-freight margin sits in the lower part of that range. The freight-and-insurance wedge between an f.o.b. and a c.i.f. price is conventionally a few per cent (the c.i.f./f.o.b. factor applied in trade statistics is about 1.06); recorded c.i.f./f.o.b. margins for former-Soviet economies are too noisy to use directly, and for several the recorded f.o.b. figure exceeds the c.i.f. one [Arvis et al., 2010], so we do not lean on them. Adding a wholesale distributive margin of a few points to a single-digit freight wedge gives the lower part of the Bureau of National Statistics bracket. The resources table’s trade-margin component alone (net of transport) is 48% for machinery and electronics, almost the whole 49% ceiling, which confirms transport is a small share of the full margin for this category but does not, on its own, pin a wholesale-only share: the table does not separate wholesale from retail, so it cannot independently validate a band meant to exclude the retail leg. We use **6–14%** and treat it as a calibration, not an estimate, because no line-by-line margin, and no wholesale/retail split, is identified in our data. Since $\bar{v}^{TT} \approx \bar{v}^M$ the input–output step does no work, so the headline is a direct function of m and we carry it as a band. (We do not use the matched-cell unit-value wedge to pin m either: censoring negative-margin cells inflates the gross margin to about 34%, while the value-weighted aggregate is negative, and both are artefacts of the c.i.f./f.o.b. basis mismatch and of outbound under-invoicing rather than measures of domestic value retained. We do not fully reconcile the 34% figure with our band: the under-recording bound we can support elsewhere in the data (13–21%, below) is not large enough on its own to explain the gap, so part of the 34% likely reflects the basis mismatch rather than under-invoicing alone, but we cannot separate the two with the data at hand, and we treat 34% as an upper bound the paper’s qualitative reading must survive, not a validated point estimate.) Domestic value added per dollar rerouted is then $m \bar{v}^{TT} \approx \mathbf{5–11\%}$, midpoint about **8%**, against **76%** per dollar of domestic manufacturing output, so a rerouted dollar creates on the order of a tenth

as much domestic value as a produced one. Of the \$479m in incremental exports to Russia after 2022, Kazakhstan retains roughly **\$23–53m** as domestic value added.

Because the propagation step is close to the identity, the headline is a one-for-one function of m , and the reader can substitute a different margin directly (Appendix D, Table D.1, gives the full sweep, the input–output cross-check and the matched-cell wedge detail). Sweeping m across the full Bureau of National Statistics bracket: at the transport-only end ($m \approx 1\%$) a rerouted dollar captures about one-hundredth of a produced one; across our 6–14% band, one-sixteenth to one-seventh; the “corridor, not factory” reading (a rerouted dollar worth on the order of a tenth of a produced one) holds for any m up to about 12%. It weakens to “about a fifth” only at $m \approx 19\%$, to “about a third” at $m \approx 32\%$, and reaches about a half only at $m \approx 49\%$, the full trade-and-transport margin that includes the retail leg the corridor never performs. The censored matched-cell gross margin (34%), even taken at face value, still leaves a rerouted dollar at about a third of a produced one. The qualitative conclusion is thus robust to any margin a wholesale-and-freight reading can support; it fails only if the corridor in fact captures a near-complete retail-inclusive markup. The one piece of evidence that bears directly on a markup that large is the value-weighted aggregate unit-value wedge, which is itself below one; we read that as suggestive only, because it is depressed by outbound under-invoicing and the c.i.f./f.o.b. basis mismatch.

The multipliers matter little by comparison. Even at $\bar{v}^M = 0.40$ the ratio is about one to five; recomputing $\bar{v}^{TT}$ and $\bar{v}^M$ from the Bureau of National Statistics symmetric input–output table (68 products, 2023) rather than the OECD ICIO gives $\bar{v}^{TT} = 0.89$ and $\bar{v}^M = 0.74$, moving the rerouted-vs-produced ratio from about one-in-ten to one-in-eight, still an order of magnitude. Under-invoicing moves it only so far: for domestic value added to reach one-fifth *of the gross rerouted flow* (a more demanding benchmark than one-fifth of a produced dollar, which $m \approx 19\%$ already reaches), the trade margin would have to be about 25%, which, holding the recorded import cost fixed, requires exports to Russia to be under-recorded by 13–21%. Under-recording on that scale is within the range found for transit trade [Fisman and Wei, 2004] and cannot be excluded; but even at that bound Kazakhstan retains at most about a quarter of what a producing economy captures.

6.3 Fiscal and macro

The reorientation-attributable flow (incremental exports to Russia over the pre-2022 baseline) is about \$479m cumulatively over 2022–2025 (\$107–134m per year). Customs duty on it, at an EAEU common external tariff of 4–8% on the share formally cleared in Kazakhstan, is on the order of **\$22m over four years** (about \$5m per year); net import VAT is close

to zero because the onward supply to Russia is intra-union and zero-rated. Kazakhstan’s annual customs revenue is of the order of \$4–5bn, so the reorientation adds well under 1%. At the macro level the gross reoriented flow, about \$0.6bn a year with the inbound and onward legs combined, is 0.2–0.3% of GDP; the 2022 current-account surplus, the reserve accumulation of 2023–2025 and the rise in services value added are an order of magnitude too large to be driven by it.

7 The (non-)investment response

If the demand shock were seeding domestic value added, we would expect a rise in investment in the sectors the flow passes through. We find none, though with only four post-2022 years the deal-count test can rule out only a large response, not a modest one.

Deal-making in manufacturing of tradeable goods, transport and logistics, and wholesale distribution runs at **7.6 per year** in 2015–2021 and **7.2 per year** in 2022–2025 (a Poisson rate ratio of 0.96, 95% CI 0.59–1.53). The two thinnest years are 2015, with a single recorded deal as the databases ramp up their Kazakhstan coverage, and 2022, the unrest year (Figure 4). Excluding the thin 2015 first year the pre-2022 rate is 8.7 per year, so the post period is a slight decline on either basis. Dropping the 2022 trough, 2023–2025 runs at 8.7 per year (rate ratio 1.14, 95% CI 0.69–1.86). The test has 80% power only against an increase of roughly 80% or more, so it excludes a surge in deal-making but not a modest step-up. The composition of that activity says more than the count. Reported deal *value* in these sectors rises, but it is concentrated in ownership transfers of existing or distressed assets, not new productive capacity: the ArcelorMittal Temirtau to Qarmet nationalisation (2023), Eurocopter Kazakhstan and the Lokomotiv rolling-stock plant (re-nationalisations, 2024), and a \$0.8bn 2025 consolidation of rail-freight and forwarding assets. Greenfield and new-capacity transactions average under two per year and never exceed four, with no post-2022 increase. Across the whole 2015–2025 window the data contain **no** transaction in the specific product lines that dominate the reorientation, though we do not lean on this, because the pre-2022 base rate in those narrow lines is also near zero, so the post-period zero carries little information on its own. We would have coded as a positive investment response any greenfield or capacity-expansion deal, in any of the three databases, tagged to a surge-basket HS6 *or* to its parent electrical-machinery, components or instrument-manufacturing class. On that rule the count is zero before and after, and the informative statistic is the sector-level rate test above, which rules out a step-up of about a doubling or more.

The result holds across the three databases. Table 3 reports deal counts for the same Kazakhstan universe from each source; coverage differs in level (the transaction-oriented

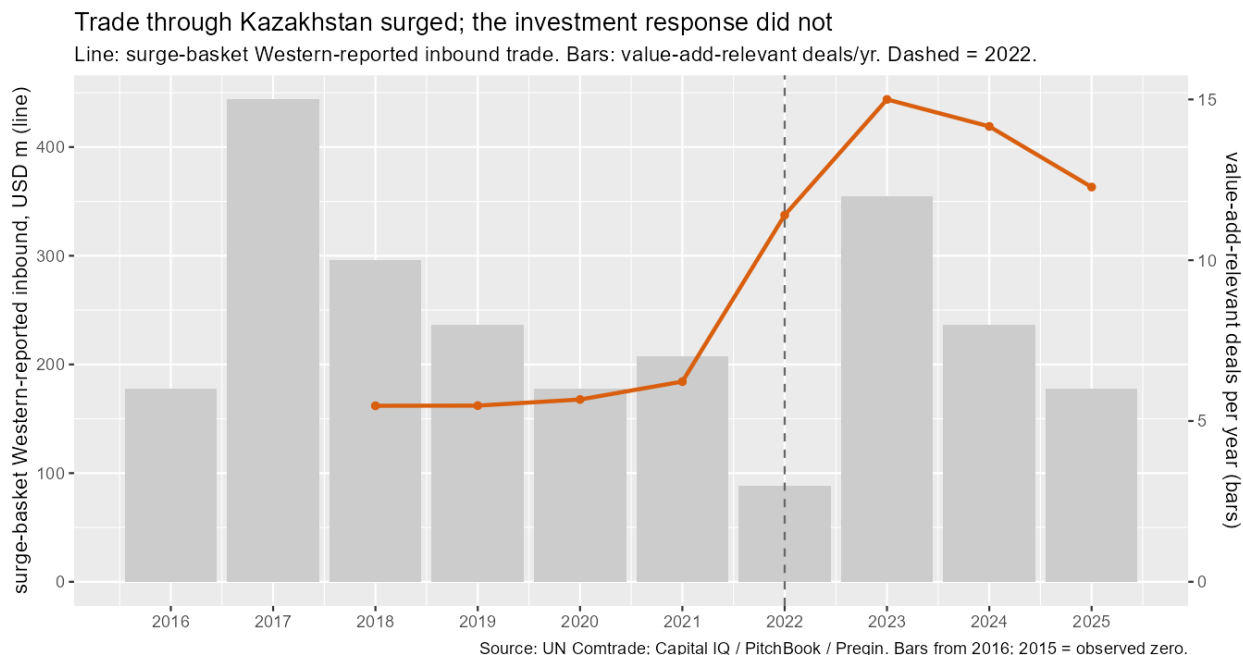


Figure 4: Trade through Kazakhstan surged; the investment response did not. Line: surge-basket *Western-reported* inbound trade (partner-reported mirror), USD million (left axis; the West + China total is several times larger but was already growing before 2022). Bars: annual count of M&A/PE deals in manufacturing of tradeable goods, transport and logistics, and wholesale distribution (right axis). Dashed line: 2022. Bars shown from 2016; 2015 has a single recorded deal and is omitted (Section 7). Source: UN Comtrade; Capital IQ/PitchBook/Preqin.

Capital IQ extract records more mid-market M&A, the venture-heavy PitchBook more small early-stage rounds, Preqin fewer of either), but none shows more than a flat rate in manufacturing or logistics deal-making (5.1 to 5.2, 2.0 to 1.5, and 0.4 to 0.5 per year). In the Capital IQ extract, which best captures the mid-market industrial deals that a genuine capacity response would generate, the count of manufacturing, transport and distribution deals is essentially unchanged before and after 2022 (5.1 versus 5.2 per year).

Kazakhstan is adding vehicle assembly (Chinese marques, KIA, Škoda) and some appliance capacity, but this predates and is largely independent of the reorientation, is kit assembly with low domestic value added, and serves the domestic and Eurasian consumer market rather than the re-export flow to Russia. The state investment corporation's disclosed post-2022 pipeline is domestic food, energy and infrastructure, not corridor processing [Qazaqstan Investment Corporation et al., 2026]. The one investment signal that is reorientation-linked, a 2025 consolidation of cross-border rail-freight and forwarding assets, deepens Kazakhstan's role as a forwarding operation, not a producing one.

8 Which gate binds?

(1)’s threshold is on a sum, not a product, so no single term of (2) is dispositive on its own; what follows states, for Kazakhstan, what each argument in Sections 5–8 establishes on its own terms. The market-access gate is open: under the Eurasian Economic Union the good moves to Russia with no transformation and no border, so V_T is large. Two pieces of evidence are suggestive, not dispositive (Section 3), that a domestic plant’s transformation margin sits below the thin re-export markup ($\mu_P \lesssim \mu_T$): the sector’s 0.69 value-added multiplier, which bears on μ_P only loosely, and its near-zero domestic base, which is itself the outcome the paper is trying to explain and so risks circularity as evidence for the margin comparison. If that comparison holds, building is dominated in (1) before the option value is added; if it does not, the null is equally consistent with Kazakhstan simply lacking a comparative advantage in the sector, customs union or not (Section 3). The re-export flow is not distinguishable in price or weight from ordinary Kazakh trade (Section 6). An open market-access gate is close to sufficient for $R \approx 0$ only jointly with that margin condition, not on its own, which limits what the case can say about the other two gates: we observe a single configuration, with no gate clearly favourable and the three not separately identified. What we can add is a state-fund pipeline that is inconsistent with the institutional gate as the binding constraint for this null (the least capital-constrained investor in the economy did not build), and a durable-versus-uncertain shock comparison that is suggestive of an adverse irreversibility gate but confounded. Neither identifies a gate cleanly, and we set out the limits of each below.

Kazakhstan’s state investment corporation (QIC, a subsidiary of the Baiterek group) faces neither an external-finance nor an exit constraint, so the institutional gate does not bind for it the way it binds for an outside investor. Its deployment in fact *rose* sharply after 2022: about \$2.2bn cumulative over 2015–2025, with about \$1.5bn of it deployed in 2025 alone, more than twelve times the \$120m deployed in 2024, of which \$1.0bn was a single convertible-bond investment in the steel producer Qarmet [Qazaqstan Investment Corporation et al., 2026]. But the report’s named QIC-linked investments are a minority stake in Almaty International Airport, a 160MW power plant in Mangystau, a poultry producer, and the Qarmet bond: domestic infrastructure, energy and food, none in a surge-basket line and none in corridor logistics built for the flow. The report does note that manufacturing became a prominent sector in the market’s recorded deal activity from 2023 onward, but that finding covers the market as a whole together with QIC, not these specific named investments. This is illustrative rather than a controlled test (the report gives no pre-2022 baseline and no sector-level denominator), but the least capital-constrained investor

in the economy did not build for the reorientation either.

Over the same window, and inside the same institutions, Kazakhstan’s *durable* passenger-vehicle demand shock (Western marques withdrawing from Russia, a domestic consumption boom, and Eurasian market access) was met with real capacity: a combined Changan/Haval/Chery assembly facility of about 90,000 units, a KIA plant (about \$200m), local assembly of several Škoda models, and a car-multimedia plant in Almaty (2024). The re-export shock was met with no plant at all. The comparison is illustrative, not a controlled test. In the deal databases auto-sector transactions actually *fell* after 2022 (six in 2015–2021, three in 2022–2025); the capacity evidence comes from plant announcements and public reporting. A parallel announcement search for the same 2022–2025 window, using the same public-reporting method, on electrical-machinery, component and instrument capacity in Kazakhstan turned up no comparable project; general reporting on the sector describes market growth and import volumes but no named new-capacity announcement in the reoriented lines (Appendix F). We treat this as a weaker check than the auto-sector search, which drew on more extensive public reporting on that sector specifically, rather than as an equally thorough symmetric search, but it does not overturn the null and is one more piece of evidence for it. Much of the auto capacity also predates the reorientation (Section 7). The two shocks also differ along several axes at once: persistence and certainty, sunk-cost intensity, whether market access requires domestic content (EAEU local-content rules bind for vehicles but not for the re-export, so the comparison also moves the market-access gate), and the destination market. The pattern is consistent with a durable, transformation-requiring shock drawing capacity where a re-export shock does not; it is not an experiment on any single gate.

A 2022 entrant faced an uncertain shock, not an observed transitory one. The flow to Russia peaked in September 2022 and fell by more than a third over the second half of 2023, from about \$15m per month in 2023-H1 to about \$9m in 2023-H2, as trade-monitoring measures tightened (the EU’s eleventh sanctions package, Council Regulation (EU) 2023/1214 of 23 June 2023, introduced a transit-monitoring provision). But that within-year decline is a correction from the September-2022 spike to a plateau, and not a decay path: on the *annual* series exports to Russia in these lines run \$128m, \$145m, \$119m and \$133m across 2022–2025, roughly ten times the pre-2022 baseline, with no downward trend and 2025 above 2024 (Table 1; with monthly reporting suspended after February 2024 and resumed only in 2025, the sub-annual path across the 2024 transition is not observed). *Ex post* the flow persisted; what a 2022 entrant faced was high *uncertainty* about its duration. The flow is policy-contingent (successive sanctions packages, monitoring provisions, compliance exposure) and its duration was unknowable. In the framework’s terms this is a high- σ configuration, which raises Ω and delivers the same null as a low- ρ one; we do not claim the realised persistence

was low.

The state fund’s pipeline is inconsistent with the institutional gate for the *shock-specific* null, but there is one place in the case where a single gate varies while the others hold, and it points to institutions: the durable adjacent opportunities of Section 10, for which the irreversibility gate is plausibly open (warehousing and transport support, with a domestic value-added multiplier of 0.82; machinery and fabricated-metal import substitution), are also financed only slowly and only by directed state capital. An open irreversibility gate with no private investment response is evidence that the institutional gate binds *there*. The domestic private-equity market’s exits run overwhelmingly through privately negotiated routes rather than public markets: trade sale, buyback and secondary sale together account for about three-quarters of surveyed exit preferences, against under a tenth for a public listing [Qazaqstan Investment Corporation et al., 2026]; capital is dominated by state and development-finance money, and the 2022–2025 pipeline is crowded by nationalisations. This is the institutional-voids configuration of Khanna and Palepu [2000] and Rajan and Zingales [1998], and it is the subject of a companion paper (available from the author).

9 Moderators and external validity

The decomposition organises seven moderators, listed in Table 4 with the value each takes for Kazakhstan. Its empirical content is not in the functional form of (2), which we do not derive and do not treat as a necessity result, but in the case-by-case arguments of Sections 5–8: R is plausibly higher when the market-access gate is closed to a pure re-export and the irreversibility gate is open, and higher still when the institutional gate is open as well. A country unfavourable on the first two is unlikely to show a response irrespective of its institutions, and a country favourable on all three is a good candidate for a supply response even with a shallow capital market.

Within the reorientation group, Armenia and the Kyrgyz Republic received the same shock, with surge-basket exports to Russia rising several-fold from 2022 (Armenia on the same break timing as Kazakhstan; the Kyrgyz Republic elevated from 2022 on a noisier series – Appendix E), and they share Kazakhstan’s value on every moderator: the shock is event-driven, its persistence uncertain *ex ante*, policy risk is high, and the customs union removes the need to transform. Two things are worth noting. Both neighbours’ flows had already receded by 2024–2025 (Armenia’s surge-basket exports to Russia: \$71m, \$93m, \$42m, \$13m over 2022–2025; the Kyrgyz Republic’s \$12m, \$30m, \$12m, \$12m), whereas Kazakhstan’s persist (\$128m, \$145m, \$119m, \$133m). The neighbour reversion shows the flow could recede, and the Kazakhstan asymmetry means the “same shock” claim holds for

the 2022–2023 surge but not for its persistence. We do not, however, have deal-level data for Armenia or the Kyrgyz Republic (our commercial extracts are Kazakhstan-only), so we cannot test the investment null for them directly; what we can say is that neither has a domestic component-manufacturing sector to expand.

The named population of post-2022 intermediaries is {Kazakhstan, Armenia, the Kyrgyz Republic, Belarus} inside the Eurasian Economic Union and {Georgia, Türkiye} outside it. The distinction matters for the framework: non-members route goods to Russia across a customs border, so their market-access gate is not fully open, and the framework predicts they are the more likely of the two groups to see a domestic processing or assembly response. On the trade side the shock is visible in the non-members too: Türkiye’s surge-basket exports to Russia roughly double at the 2022 break (from about \$58m in 2021 to \$119m in 2022 and \$207m in 2023; sup- F of 11.9, $p = 0.007$), while Georgia barely participates in this particular flow (under \$2m a year throughout, no clean break); Appendix E gives the full series and break tests, alongside Armenia and the Kyrgyz Republic. We cannot run the deal-database investment test for either: our commercial extracts are Kazakhstan-only. As a lower-fidelity substitute we ran a public-record announcement search, on the same 2022–2025 window and the same electrical-machinery, component and instrument categories as the corridor’s own surge basket, for new capacity in Türkiye and Georgia (Appendix F gives the protocol); it turned up no comparable project in either country. This does not confirm the framework’s prediction that the non-members would be more likely to see a processing or assembly response: a public-record search is a weak instrument for detecting a marginal capacity change, especially in Türkiye’s large pre-existing manufacturing base, so absence of a reported project is not evidence of absence in the way the Kazakhstan-focused commercial-database test is. The prediction remains untested at the standard the rest of the paper holds itself to, and a proper test would need the same kind of greenfield-investment register (fDi Markets or Orbis Cross-Border Investment) named in the Limitations below. The external-validity claim we can support is therefore narrow: the *demand shock* generalises across the intermediary population, while the *investment null* is established only for Kazakhstan, where the market-access gate is fully open and a domestic component sector barely exists.

As motivating contrast cases, economies that drew capacity investment from a post-2020 tradeable demand shock did so with a favourable configuration on the first two gates, not necessarily the third. Vietnam and Mexico received China-plus-one and nearshoring reallocations widely read as *structural* [e.g. Juhász et al., 2024], and in autos the tightening of rules of origin under the United States–Mexico–Canada Agreement made domestic content a condition of market access (a closed market-access gate that forces $V_T \rightarrow 0$). Both invested in capacity despite financial markets that are not uniformly deep. This is the pattern the

framework predicts: persistence and a transformation requirement do the work, and financial development scales the response.

Firm-level evidence makes the same point at smaller scale: demand shocks induce upgrading when meeting the demand requires transformation, not when trade is a costless substitute. A randomised expansion of foreign market access for Egyptian rug producers, who could not resell into that demand without meeting buyers' quality specifications, raised profits, productivity and product quality [Atkin et al., 2017]; the 1994 Mexican peso devaluation raised the return to exporting for Mexican plants that could not serve the newly cheaper-to-reach foreign market without upgrading quality and, with it, the skill mix of their workforce [Verhoogen, 2008]. In both cases V_T is effectively unavailable – the demand cannot be captured by pure re-export or resale – so the shock's effect passes through the market-access gate into μ_P and I rather than being absorbed as V_T , the opposite of the Kazakhstan configuration.

We cannot, however, separate the irreversibility and institutional gates across countries. Doing so would require regressing sector-level greenfield investment on shock persistence and financial depth in a panel of intermediaries and contrast cases. In the sample available to us the two are collinear: Kazakhstan, Armenia, the Kyrgyz Republic and the other Caucasus and Central Asian intermediaries are all financially shallow (private credit plus market capitalisation of 24–66% of GDP, all below the comparator median), and Türkiye is financially deeper but has too large a pre-existing manufacturing base for a marginal-capacity response to register. Country-level macro aggregates are too noisy to isolate a response of this size, and in places run the other way: Armenia's gross fixed capital formation rose about three points of GDP after 2022. The argument leans on the within-country comparisons of Section 8; a cross-country panel using greenfield-project data by sector is the natural extension.

10 Where investment would raise value capture

The reorientation is nonetheless informative for investment policy, because the trade data reveal which products have proven demand and proven routes through Kazakhstan. Table 5 sets three quantities side by side for the candidate sectors, namely the domestic value-added multiplier, the import-substitution gap (imports relative to domestic output), and the post-2022 growth in import demand, after removing the surge-basket lines from the electronics-adjacent chapters; we read them jointly, and use a simple combined score only to order the candidates, not to rate them (Figure 5).

Two groups stand out. The **logistics platform** (warehousing and transport support, and



Figure 5: Where investment buys the most domestic value added: domestic value-added multiplier against import-substitution headroom (imports 2023 / domestic output, log scale), bubble size = domestic output. The surge-basket products (computer/electronic/optical) sit at the bottom.

wholesale-trade infrastructure) has among the highest domestic value-added multipliers in the economy (0.82) and is the corridor's binding constraint, given the high and unpredictable overland logistics costs a landlocked transit economy faces [Arvis et al., 2010]; investment here creates domestic value and holds it whether or not the Russia flow persists. **Import substitution** in electrical equipment, machinery, fabricated metal products and chemicals is the second: Kazakhstan imports six to eight times what it produces in electrical equipment and machinery and roughly twice its output in the other two; the domestic value-added multiplier is around 0.80, and a nascent deal base exists. By contrast, localising the surge-basket products is the worst option: computer, electronic and optical manufacturing has among the lowest value-added multipliers in the economy (0.69, tied with motor-vehicle kit assembly) and a near-zero domestic base.

These opportunities are investable at scale only if the state investment corporation shifts from lead direct investor to co-investor behind private leads, with larger syndicated tickets and a credible listing-based exit, rather than relying only on the privately negotiated trade sales, buybacks and secondary sales that dominate exit preferences today. That is a question

about capital-market institutions, taken up in a companion paper.

11 Policy implications

The framework of Section 3 is also a menu of levers. A demand shock of this kind industrialises a transit economy only if at least one of three things holds: the market-access gate is closed to a pure re-export, the irreversibility gate is opened, or the institutional gate is opened. Each is taken in turn below for Kazakhstan, followed by what its position on the China–Europe land route adds and what its principal supplier could do. The discussion is normative and forward-looking, separate from the paper’s positive results, and rests on the framework, not on new estimation.

11.1 Kazakhstan: raising the investment response

Market access. The customs union makes trade a near-perfect substitute for production, which is close to sufficient for a null only jointly with a small transformation margin, not on its own (Section 3). The corresponding lever is to make some domestic value added a condition of capturing the flow: procurement and investment-incentive rules that reward domestic processing, and use of the customs-union position to attract assembly and not only transit. The target is the substitution and platform sectors and the *next* shock of this kind, not the surge-basket goods themselves (the “Avoid” rows of Table 5, where domestic assembly of imported kits earns a low multiplier of 0.69 and forcing it would mostly deter the throughput): a standing rule that flows above a threshold trigger a domestic-content review would channel investment toward the 0.80-multiplier lines and away from low-value final assembly. This is the sharpest lever and the hardest. It trades some current throughput for the chance of durable capacity, and a unilateral local-content requirement on *onward* trade sits uneasily with the Eurasian Economic Union’s common-market rules on the free movement of goods [Eurasian Economic Union, 2014], so the feasible form is a domestic incentive or procurement preference, or a measure agreed at union level, and not a border restriction Kazakhstan can impose alone.

Irreversibility. Building beats waiting only when the shock is durable and certain enough and the technology cheap enough to sink. Policy can move two of these three. It can lower the effective uncertainty σ with multi-year commitments and credible demand signals: anchor and offtake contracts, state-backed demand aggregation across ministries and state firms, and a stable rule for how long an incentive lasts. It can lower the effective sunk cost I with shared pilot-line and testing infrastructure, ready-built special-economic-zone

tenancy, and co-financing of first-of-kind facilities so that no single entrant carries the whole irreversible bet. The tension in this lever is fiscal risk-shifting: a state-backed offtake or demand-aggregation contract moves the demand risk from the investor to the budget, and if the flow stops the loss falls on the taxpayer rather than the firm. Given the 2022–2025 nationalisation wave, which already concentrated industrial risk on the state balance sheet, the defensible version is a capped, time-limited and partial guarantee tied to domestic value-added milestones, not an open-ended purchase commitment.

Institutions. The durable adjacent opportunities the shock reveals (logistics and transport support, machinery and fabricated-metal import substitution) are financed only slowly and only by directed state capital. The lever is to move the state investment corporation from lead direct investor to anchor limited partner and co-investor behind independent managers, with larger syndicated tickets and a credible public-market exit route: a public listing draws under a tenth of surveyed exit preferences today, against the trade sales, buybacks and secondary sales that together dominate [Qazaqstan Investment Corporation et al., 2026], and building the domestic private-equity exit market alongside it means widening that route.

Across all three levers the first call on investment is the logistics platform itself (warehousing, transport support and Trans-Caspian capacity): as Section 10 shows, it carries the highest domestic value-added multiplier of the candidate sectors and creates value that holds whether or not any particular flow persists.

11.2 Corridor leverage in the current geography

Kazakhstan’s bargaining position on these questions is stronger than the value-capture numbers alone suggest. The closure of the northern land corridor and the 2023–2025 disruption of the main maritime routes, described in Section 2, leave the shortest China–Europe land path that avoids both Russia and the maritime chokepoints running through Kazakhstan and across the Caspian. Kazakhstan is close to an unavoidable node on that route, a structural bargaining chip with China, which wants route diversification and resilience, and with the European Union, which wants a land option that does not run through Russia.

The long-run play, in the framework’s terms, is to use corridor access, capacity and reliability guarantees as the lever the customs union otherwise removes: to bundle throughput and priority-handling commitments with requirements or incentives for co-located processing, local content, joint-venture assembly, and standards and technology transfer, converting transit rent into value-added capacity rather than collecting fees for pure pass-through. It is the same recommendation as the market-access lever above, reached from the geography rather than from the customs-union rule.

The value here is in the option, not the transit fee. The Trans-Caspian route carried on the order of 2–3 million tonnes in 2023–2024 and is targeted at roughly 10 million by 2030 [World Bank, 2023], against China–EU merchandise trade of some \$800bn a year; transit and handling charges on even the 2030 volume are a fraction of one per cent of Kazakh GDP. The route matters as the marginal land link China and the EU would need if the maritime disruptions persisted, and its current revenue is beside the point.

The leverage is bounded. Transit demand is contestable: a southern route through Iran and Türkiye and competing Caspian nodes in Azerbaijan and Georgia would capture part of any rent. Caspian ferry and port throughput and the rail break-of-gauge cap how much traffic the route can actually take. It also lasts only as long as the chokepoint disruptions persist and China’s route diversification remains incomplete.

11.3 China: lowering barriers to greater Kazakh value capture

Because China supplies about two-thirds of the inbound flow, some of the levers sit on the Chinese side. These are hypotheses the paper cannot test, offered as directions rather than findings. Value-added-tax rebate and export-processing rules, major instruments of Chinese industrial policy whose rates vary sharply by product and demonstrably shift export composition [Gourdon et al., 2022], currently favour finished-goods export over component export plus assembly in Kazakhstan, tilting firms toward shipping the completed article rather than locating a stage of production on the route. Approval friction for mid-sized outward-investment joint ventures in Central Asia raises the fixed cost of the kind of plant that would add domestic value. Mutual recognition of standards and certification would let a Kazakhstan-assembled good qualify for both the Chinese and the Eurasian market without a second conformity process. Coordination of bonded-processing zones under the Trans-Caspian and Belt-and-Road frameworks would give co-located assembly the same customs treatment as pure transit.

None of these levers substitutes for the others: the market-access and corridor-leverage levers are the most powerful and the hardest to pull, and the institutional lever is a precondition for private capital to act once those two have been pulled.

12 Discussion

Kazakhstan’s trade with Russia and with the West in these products moved by an order of magnitude after 2022. The domestic economic content of that movement is small: value added about a tenth of the manufacturing equivalent, customs duty under one per cent of

revenue, a fraction of one per cent of GDP, and no detectable investment response. The country is functioning as a corridor, not a site of production.

The result speaks to a general question: when does a trade shock industrialise a country, and when does it merely pass through? Production is a more plausible response the more that the demand cannot be met by trade alone (a closed market-access gate), the more that the shock is durable and certain enough, and the technology cheap enough to sink, that building beats waiting (an open irreversibility gate), and the more that the capital market can fund the projects that pass (an open institutional gate); we treat these as three considerations that raise the odds of a supply response, not as conditions we derive to be jointly necessary. The Kazakhstan case pins down the first condition and half of the third: the customs union makes trade a close substitute for production, so the shock is absorbed into the trade channel, and the state investment corporation, which does not face the capital-market constraint, did not build either, which is inconsistent with the institutional gate as the binding constraint for this null (though its mandate and its cross-border-compliance exposure cloud the comparison). Whether the irreversibility gate *independently* binds the case cannot say: because the re-export flow in fact persisted through 2025, the null cannot be read as a response to an observed transitory shock; what a 2022 entrant faced was uncertainty about the shock's duration, and the durable-versus-uncertain comparison moves more than one gate at a time. The institutional gate explains the missed alternative: the adjacent durable opportunities the shock reveals, in logistics and machinery, are left to slow, directed state capital and go unfunded by private capital even though their irreversibility gate is open. Armenia and Türkiye show the same 2022 surge (the Kyrgyz Republic and Georgia are elevated from 2022 on noisier series; Appendix E) and, on the evidence we have, no domestic component-manufacturing sector to expand; the framework's contrast cases (structural reallocations, or rules-of-origin regimes that force domestic content) are where a supply response does occur, even without a deep capital market. The static value Kazakhstan misses (value added a tenth of the manufacturing equivalent) is the smaller part of the cost; the larger part is dynamic. A rules-of-origin or local-content regime that forced a stage of production onshore would also create the backward linkages through which foreign suppliers transfer technology and raise the productivity of domestic firms [Javorcik, 2004], and the supplier learning and diffusion that come with a durable production relationship [de Souza et al., 2024]. A pure corridor forgoes all of this: the goods pass through without a domestic firm ever entering the supply chain, so there is no linkage to learn from. That is where the long-run cost of being a corridor sits, and it is not captured by the value-added arithmetic.

A contemporaneous manufacturing expansion, and why it is not a counterexample. Over the same window Kazakhstan’s aggregate manufacturing sector did grow: for January–July 2026 it accounted for about 47% of industrial output against 46% for mining (the second year manufacturing has run ahead), with manufacturing up about 9% year on year while mining fell about 4% and crude-oil output about 9% [Bureau of National Statistics, Agency for Strategic Planning and Reforms of the Republic of Kazakhstan, 2026]. Read against this paper, the expansion is consistent with the framework. Part of the crossover is the denominator: mining and oil contracted, so a rising manufacturing *share* overstates the real divergence. The growth is concentrated in passenger vehicles and assembly (machinery output up about 22%, on full-cycle plants with roughly 190,000 units of announced capacity) and in domestic-demand import substitution (fabricated metals, chemicals, pharmaceuticals), and not in the re-export lines of this study, where no capacity was added. The mechanism, too, is the one the framework names: minimum local-content shares written into subsoil-user contracts, a shift to mandatory procurement from domestic producers by large firms and state agencies, off-take contracts underwriting demand, and directed state and development finance behind new plants [Government of the Republic of Kazakhstan, 2026, Eurasian Economic Union, 2014]. That is a deliberately *closed* market-access gate plus directed institutions, exactly the configuration under which the framework predicts a supply response. Where Kazakhstan required domestic content or supplied the demand itself, investment followed; where the good could move onward untransformed inside the customs union, it did not.

Limitations. Four limitations are central. First, the case observes a single configuration, in which the three gates are not separately identified. It can still identify the market-access gate as open and find evidence inconsistent with the capital-market institutional gate as the binding constraint for the shock-specific null; what it cannot establish is that irreversibility independently binds. The durable-versus-uncertain shock comparison and the state-fund pipeline are suggestive corroboration, not clean experiments, and the cross-country panel that would separate the irreversibility and institutional gates requires greenfield-project data by sector and country. Second, the value-capture headline is a calibration: because the input–output step does almost no work, it scales one-for-one with a trade margin we choose rather than estimate (Section 6); the “corridor, not factory” reading (a rerouted dollar worth about a tenth of a produced one) holds for a margin up to about an eighth, weakening to “about a fifth” at a margin near a fifth. Third, the investment null is *over*-determined, since the January 2022 unrest, the cross-border compliance exposure noted in Section 2, and the nationalisation wave would each depress entry, so we bound its magnitude (Section 7) rather

than attribute it to the trade shock alone. Fourth, the deal databases capture *transacted* investment (M&A and private financing rounds) and under-cover pure greenfield capacity that is built without an external transaction, and they are Kazakhstan-only, so we cannot use them to test the framework’s prediction that non-EAEU intermediaries (Türkiye, Georgia) should be more likely to see a processing response. A dedicated greenfield register (fDi Markets, or Orbis Cross-border Investment) would close both gaps; the plant-level evidence we use in Section 8, and the public-record searches in Sections 8 and 9 for components-sector capacity in Kazakhstan and for a differential response in Türkiye and Georgia, are manual substitutes, weaker instruments than a purchased register. The two searches are not evidence of the same kind: the Kazakhstan components search finds no capacity where the framework predicts none, which is consistent with the framework; the Türkiye/Georgia search finds no differential response where the framework predicts one *would* appear for non-EAEU intermediaries, which is a weak-instrument null that fails to find the predicted difference and is therefore not supportive of the prediction, not consistent with it. The cross-country prediction remains untested at the standard the rest of the paper holds itself to.

Several measurement choices also bear on the estimates. The inbound side is partner-reported (mirror) data, which overstate Kazakh absorption for onward-moving goods. The inbound flow is dominated by a large pre-existing China component, so we measure value capture on the incremental Western flow and read the West + China inbound only at the product level, where line and year fixed effects absorb the China trend. The surge basket is selected on the post-2022 outcome; the externally compiled priority list breaks at the same date, and its significant coefficient is carried substantially, though not entirely, by the 24 lines it shares with the surge basket – the 26 it does not share are, against a control purged of the surge basket, independently significant on the outbound flow but at a smaller magnitude (Table 2), so this is partial, smaller-scale corroboration rather than a fully independent replication. We therefore read the cross-sectional difference-in-differences alongside the structural breaks and the neighbour comparison rather than on its own, and a placebo on the largest civilian lines is significant with the opposite sign, indicating the control pool is not homogeneous in trend. The retained margin is taken from the national-accounts convention (6–14%), not estimated line by line, and the value-added multipliers come from the OECD ICIO (2019), cross-checked against the Kazakhstan national input–output table (2023) with the same qualitative result.

A Deal-database query specification

For online publication.

The Kazakhstan deal universe is extracted from three commercial databases (Capital IQ, PitchBook, Preqin). Common parameters: *target/issuer geography* = Kazakhstan (country of incorporation or primary operations); *announced date* between 1 January 2015 and 31 December 2025; *deal types* = merger/acquisition, majority/minority stake, private-equity buyout and growth, venture (all stages), joint venture, and PIPE/private placement; *status* = completed or announced (pending and withdrawn recorded separately). Financials in USD at announcement.

Capital IQ (S&P): Screening > Transactions; Target Company Location = Kazakhstan; Transaction Types as above; Announced Date range as above. Export all columns; extraction date recorded in the replication file.

PitchBook: Advanced Search > Deals; Company HQ Location = Kazakhstan; Deal Types = M&A, Buyout/LBO, PE Growth/Expansion, all VC stages, Angel, Accelerator/Incubator; Deal Date range as above.

Preqin: Private Equity > Deals & Exits; Portfolio Company Country = Kazakhstan; Deal Date range as above.

De-duplication is on target name, announced date (± 30 days) and deal type. Sector classification is harmonised to ISIC rev. 4 divisions from each database’s native taxonomy via a documented crosswalk. The three extracts, the crosswalk, and the reconciliation script are in the replication package, which pins each source’s extraction date and its earliest year of reasonably complete Kazakhstan coverage.

B Product-set construction

For online publication.

The candidate set is 75 HS6 lines: 50 from an externally compiled list of technology-intensive items published in February 2024 (the “priority list”), grouped by its publisher into six tiers, plus 25 clearly civilian control lines. The **surge basket** is defined from the data as the lines in which the mean inbound flow (Western reporters + China) and the mean outflow to Russia *each* at least doubled from 2019–2021 to 2022–2024, with the ratio computed after adding \$10,000 to numerator and denominator, and with post-period floors

of \$0.2m (inbound) and \$0.1m (outbound). Twenty-nine lines qualify; 24 are on the priority list and 5 are control lines that also surged. Of the 75 candidates, 55 clear the outbound doubling and 41 the inbound one; requiring both is the binding screen.

C Difference-in-differences robustness

For online publication.

Table C.1 collects the full battery summarised in Section 5.3. The outcome throughout is $\ln$ of Kazakhstan’s exports to Russia; γ is the coefficient on treated $\times$ post with HS6 and year fixed effects and cluster-robust standard errors by HS6, unless noted.

D Value capture: input–output detail

For online publication.

Domestic value added per rerouted dollar is $m \bar{v}^{TT}$, against $\bar{v}^M$ per dollar of domestic manufacturing output. From the OECD ICIO Kazakhstan block, $\bar{v}^{TT} = 0.787$ (trade, transport, warehousing) and $\bar{v}^M = 0.764$ (manufacturing), so the rerouted-vs-produced ratio is $\approx 1.03 m$ and the input–output step does almost no work. Recomputing on the Kazakhstan Bureau of National Statistics symmetric input–output table (2023, 57 industries matched, output-weighted within group) gives $\bar{v}^{TT} = 0.885$ and $\bar{v}^M = 0.742$, moving the ratio from 0.10 to 0.12. The BNS resources table brackets m : the transport-only margin on machinery and electronics is about 1% of basic-price value and the full trade-and-transport margin (which carries the retail leg the corridor never performs) about 49%; the trade-margin component alone (net of transport) is 48% for the same category, confirming that transport is a small part of the full margin here, though the table does not separate wholesale from retail and so cannot independently validate a wholesale-only reading. The 6–14% band is a wholesale-and-freight reading, above the floor and well below the ceiling; see below for how it relates to the 0.34 matched-cell estimate.

We do not fully reconcile the censored matched-cell gross margin (0.34) with the 6–14% band. The under-recording of outbound exports that would be needed to explain a one-fifth domestic-value reading from the gross rerouted flow is 13–21% (Section 6); that bound is not large enough on its own to bring 0.34 down into our band, so at least part of the gap likely reflects the c.i.f./f.o.b. basis mismatch rather than under-invoicing alone. We cannot separate the two components with the data at hand, and we treat 0.34 as an upper bound

the paper’s qualitative reading must survive (Table D.1 shows it does, at “about a third”), not as a validated point estimate of m .

The matched-cell unit-value evidence (annual, $n = 112$ HS6×period cells) is descriptive only. The median re-export/import ratio is 0.73 on a c.i.f./f.o.b. basis and 1.6 on a like-for-like f.o.b. basis; the civilian control basket gives 0.59 and 0.94. The pass-through slope $\log(uv_{\text{expRU}})$ on $\log(uv_{\text{mirWC}})$ is 0.48 ($p = 0.03$), and 0.20 ($p = 0.46$) against the Kazakh-reported import price. The f.o.b. wedge by tier is 0.47 (Tier 1), 3.83 (2), 1.82 (3A), 1.09 (3B), 3.94 (4A), 1.35 (4B), against 1.33 for the control basket: the wedge is uneven across tiers, and its high values sit on about half the matched gross flow, the pattern of a scarcity markup rather than an even processing margin. The cell-level weight ratio (re-exported over imported tonnage) has median 0.08 and aggregate 0.11; the aggregate weight ratio equals the aggregate value flow-through (0.11), so a sub-one weight ratio is largely a flow-through fact. Eleven per cent of cells exceed 1.05, on an interquartile range of 0.02 to 0.49. In the six near-pure-transit cells (value flow-through in $[0.8, 1.25]$) the weight ratio is 0.74 at the median (0.26 and 0.94 at the quartiles).

E Neighbour and comparator structural breaks

For online publication.

Table E.1 reports surge-basket exports to Russia for the other post-2022 intermediaries and a mean-shift (sup- F) break test on the annual asinh series, 2018–2025. Armenia and the Kyrgyz Republic are inside the Eurasian Economic Union with Russia; Georgia and Türkiye route across a customs border. Belarus is also in the Eurasian Economic Union and in the intermediary population, but it stopped reporting to UN Comtrade in 2022, so no post-2022 series is available and it is omitted from the table. The 2022 break appears regardless of customs-union status; the demand shock generalises across the population, while the investment test is Kazakhstan-only for want of deal data elsewhere. Persistence, however, does not: Türkiye’s flow peaks in 2023 at \$207m and recedes to \$73m by 2025, a reversion pattern like Armenia’s and the Kyrgyz Republic’s rather than Kazakhstan’s sustained \$119–145m; Kazakhstan is the only intermediary in the table whose post-2022 level does not recede toward its pre-2022 trajectory.

F Public-record search protocols

For online publication.

Two public-record searches are referenced in Sections 7 and 9 as lower-fidelity substitutes for a purchased greenfield-investment register (fDi Markets or Orbis Cross-Border Investment), which is unavailable to this project. Both used the same method: general web search and news aggregation for named capacity announcements (plant openings, assembly lines, production-capacity expansions) in the stated sector and window, cross-checked against the deal databases of Section 4 where a transaction existed. Neither search used a purchased press-clipping service or a systematic industry-directory sweep, so both are weaker instruments than the auto-sector triangulation in Section 7, which drew on more extensive public reporting specific to that sector.

- **Kazakhstan components search** (Section 7): window 2022–2025; sectors electrical machinery, components, and instruments (the surge-basket adjacent categories); search conducted 2026-09 using general web search for named new-capacity announcements. Result: no named new-capacity project found; general reporting on the sector describes market growth and import volumes only.
- **Türkiye/Georgia components search** (Section 9): same window and sector definitions, applied to Türkiye and Georgia; search conducted 2026-09. Result: no comparable project found in either country. As Section 9 states, this is read as an uninformative weak-instrument null for the framework’s cross-country prediction, not as a confirmation.

Neither search was pre-registered or independently replicated; both are disclosed as best-effort substitutes, not as a systematic literature or press-record audit.

Declaration of competing interest

The author is employed by the Astana International Financial Centre (AIFC), which operates the Astana International Exchange and is institutionally adjacent to Kazakhstan’s state investment vehicles, and previously worked with a consulting team on a Kazakhstan trade-strategy engagement. The state-fund investment figures used here are drawn from a report co-published by the AIFC with the Qazaqstan Investment Corporation (QIC) and the International Finance Corporation (IFC). Neither role enters the empirical analysis, which rests on public data and on commercial deal databases available under academic licence; the policy recommendations in Section 11 are the author’s own and do not represent the AIFC, QIC or the IFC.

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Data availability

The trade analysis uses public data: UN Comtrade (HS6 annual and monthly), World Bank World Development Indicators, the OECD Inter-Country Input–Output tables (2023 edition), and the Kazakhstan Bureau of National Statistics symmetric input–output and resources tables. The deal-level data are licensed from Capital IQ, PitchBook and Preqin and cannot be redistributed, but the full query specification (Appendix A) and the native source deal identifiers are in the replication package, so the extract is reconstructible by a licence holder. State-fund investment aggregates are from the published QIC/AIFC/IFC report. Code and query specifications are in the replication package, under `scripts/R/kz_passthrough/` and `scripts/R/kz_valueadd/`.

Declaration of generative AI in the writing process

During the preparation of this work the author used Claude (Anthropic) to draft and edit prose, to help write and debug the analysis code, and for structured internal review of the manuscript. All data choices, analytical decisions, results and interpretations are the author’s own; the author reviewed and edited all output and takes full responsibility for the content of the publication.

CRedit authorship contribution statement

Zhanbolat Kakishev: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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Table 2: Difference-in-differences: γ on treated $\times$ post in $\text{asinh}(y_{it}) = \gamma(\text{treated}_i \times \text{post}_t) + \mu_i + \lambda_t + \varepsilon_{it}$, HS6 and year fixed effects, cluster-robust standard errors (by HS6) in parentheses. Surge-basket and priority-list rows have $N = 600$ (75 lines $\times$ 8 years); the placebo row has $N = 200$ (the 25 civilian lines). The residual-list row is estimated on a purged sample, $N = 368$ (the 26 residual lines plus the 20-line civilian control excluded of the surge basket itself; JIE round-2 review, Referee B N1 – the full-panel control used in an earlier draft of this row double-counted the 29 surge-basket lines as "control," which is corrected here). The surge basket is selected on the inbound (West + China) and outbound flows; the priority list (50 codes) is externally compiled. Row 1 (surge basket) and row 2 (priority list) are estimated against the full 75-line panel; row 3 (residual list) is estimated against the 46-line purged sample described above, so its control pool differs from rows 1–2. No stars are reported in row 1: the surge-basket coefficient’s magnitude is not separable from the sample selection (Panel C), including for the KZ-reported-imports column, whose apparent significance should be read against Panel C’s randomisation-inference null for that basket rather than taken at face value. The residual row’s star reflects the wild-cluster bootstrap p , the reliable statistic at 46 clusters.

Treated set	Exports to Russia	Inbound (W+China)	Inbound (Western)	KZ-reported imports
<i>Panel A. asinh DiD, γ (cluster-robust s.e. by HS6)</i>				
Surge basket (29 HS6)	2.44 (0.96)	2.10 (1.06)	1.73 (0.98)	1.88 (0.63)
Priority list, external (50 HS6)	2.88 (0.74)***	2.12 (0.71)**	1.93 (0.67)**	2.94 (0.53)***
of which, 26 lines, purged civilian control	1.94 (0.90) [†]	1.12 (0.83)	—	—
Placebo (largest civilian lines)	—	−0.95 (0.25)**	—	—
<i>Panel B. Poisson (PPML) DiD in levels, e^β (multiplicative)</i>				
Surge basket (29 HS6)	$3.5 \times^*$	$2.4 \times^{***}$	—	—
<i>Panel C. selection-rule-matched and random-basket randomisation inference, exports to Russia</i>				
surge basket, rule-matched null γ : mean 2.79, s.d. 1.50; $P(\gamma^{\text{perm}} \geq 2.44) = 0.58$				
surge basket, existence null: $P(\text{rule selects} \geq 29 \text{ lines} \mid H_0) = 0.000$				
priority list, random-50-line-basket null γ : mean 0.00, s.d. 0.97; $P(\gamma^{\text{perm}} \geq 2.88) = 0.002$				

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (cluster-robust; before multiple-testing adjustment); [†] wild-cluster bootstrap $p < 0.05$, the reliable statistic at 46 clusters (analytic cluster-robust p is unreliable at this cluster count). After Holm adjustment across the four surge-basket outcomes, exports to Russia and KZ-reported imports remain significant at 5%. Surge-basket and priority-list rows in Panel A have $N = 600$ (75 lines $\times$ 8 years); the residual-list row has $N = 368$ (46 lines $\times$ 8 years, purged of the surge basket on both treatment and control sides); the placebo row has $N = 200$. The residual-list row’s wild cluster bootstrap $p = 0.035$ outbound, $p = 0.238$ inbound. Panel B: exact PPML coefficients $\beta = 1.26$ ($p = 0.02$) outbound and $\beta = 0.85$ ($p < 10^{-4}$) inbound. Panel C: the surge-basket coefficient’s *magnitude* is not separable from the post-outcome selection, but the *existence* of a 29-line basket is not a chance artefact under either a free or a trend-preserving permutation; the priority list’s coefficient is, by contrast, unusual relative to a random 50-line basket of any HS6 lines, which is consistent with it being carried substantially, though not entirely, by the (genuinely treated) surge lines it contains: the residual 26 lines’ own smaller, purged-control DiD is independently significant on the outbound flow (row 3).

Table 3: Deal counts for the Kazakhstan universe, by source and period. Rates are per year over a 7-year pre-window (2015–21) and a 4-year post-window (2022–25). The consolidated non-duplicate universe is $N = 493$ transactions; the manufacturing, transport and distribution subset is $N = 82$ (53 pre, 29 post).

Deals per year (unless noted)	Cap. IQ	PitchBk	Preqin
All deals, 2015–21	24.0	14.6	3.7
All deals, 2022–25	23.2	24.2	1.8
Mfg + transport + distrib., 2015–21	5.1	2.0	0.4
Mfg + transport + distrib., 2022–25	5.2	1.5	0.5
Surge-basket lines, 2015–25 (total)	0 ^a	0	0

^a Two Capital IQ records carry an “electronic” industry tag (a steel-wire-rope manufacturer and a finance holding company) and are misclassifications; there are no transactions in the reorientation’s product lines.

Table 4: Moderators of the investment response, and Kazakhstan’s values

Moderator	Gate	Kazakhstan, 2022	$R \uparrow$ when
Shock persistence ρ (structural reallocation vs. event-driven)	irreversibility	flow persisted <i>ex post</i> ; <i>ex ante</i> uncertain	high
Uncertainty σ / policy risk	irreversibility	high (policy-contingent)	low
Sunk-cost intensity I of the relevant technology	irreversibility	high (component mfg)	low
Domestic transformation required to reach the market (rules of origin, local content, a border)	market access	none (customs union)	required
Financial depth / exit availability	institutions	shallow, no exit	deep
Private vs. state / directed capital mix	institutions	state-dominated	private-led
Existing productive base / absorptive capacity	institutions	near-zero (electronics output $\approx$ \$230m)	a base exists

Table 5: Sector characteristics for value-adding investment, Kazakhstan. VA mult. = domestic value-added multiplier $(v'_d L_d)_j$ from the OECD ICIO block; Output = gross output, \$m; Imp./output = 2023 imports over domestic output; Imp. growth = ratio of 2023 to 2019 import value. Dashes: not applicable (platform sectors are not import-competing) or not computed. Surge-basket HS6 are netted out of the electronics and machinery chapters. The substitution rows are the four top-ranked candidates on a combined score (value-added multiplier, import-substitution gap, import growth, deal base) out of $N = 13$ ranked tradeable sectors; two platform and two “Avoid” rows are shown alongside, so 8 of the 13 appear here and the full ranking is in the replication output.

	Sector	VA mult.	Output \$m	Imp./output	Imp. growth
Platform	Warehousing & transport support	0.82	6,800	—	—
	Wholesale & retail trade infrastr.	0.82	81,000	—	—
Substitution	Electrical equipment	0.81	1,150	5.6	1.43
	Machinery & equipment n.e.c.	0.80	1,160	7.9	1.27
	Fabricated metal products	0.81	1,850	1.8	1.22
	Chemicals	0.79	3,480	1.7	1.52
Avoid	Computer, electronic & optical (surge basket)	0.69	230	—	—
	Motor vehicles & parts (kit assembly)	0.69	1,530	6.1	2.55

Table B.1: Priority-list HS6 codes by publisher tier, and the civilian control set. † marks the 29 surge-basket lines.

Tier	HS6 codes
1	854231†, 854232, 854233†, 854239†
2	851762†, 852691†, 853221, 853224, 854800†
3A	847150, 850440, 851769, 852589†, 852910, 852990, 853669†, 853690, 854110†, 854121, 854129†, 854130, 854149†, 854151, 854159†, 854160
3B	848210, 848220†, 848230†, 848250, 880730, 901310†, 901380, 901420, 901480
4A	847180†, 848610, 848620, 848640, 853400, 854320†, 902750†, 903020, 903032†, 903039†, 903082
4B	845710†, 845811†, 845891†, 845961, 846693†
Control	090111, 170490, 190590, 210390, 210690, 220300, 330300, 330499, 340111, 392321, 392490†, 401110†, 480256, 610910†, 611020†, 620342, 630260, 640399, 691110, 732690, 841810, 870899, 940360, 950300, 961900†

Priority list: EU/US/UK/JP List of Common High Priority Items, version of February 2024. Tier labels are the publisher’s; we carry them unchanged.

Table C.1: Difference-in-differences robustness, surge basket, exports to Russia. The placebo’s fake treatment falls on the six largest civilian lines by pre-2022 size; none of the six is in the surge basket.

Specification	γ	s.e.	p
Baseline (HS6 + year FE)	2.44	0.96	0.013
wild-cluster bootstrap p (B = 1999)			0.010
Holm-adjusted across 4 surge-basket outcomes			0.039
+ size-decile $\times$ year FE	2.29		0.003
Drop 2022 entirely	2.71	1.05	0.012
Donut (drop 2022, treatment year = 2023)	2.71	1.05	0.012
Poisson (PPML) in levels, e^β	$3.5\times$		0.020
Priority list (50 HS6, externally compiled)	2.88	0.74	< 0.001
of which, 26 lines, vs. purged civilian control ($N = 368$)	1.94	0.90	0.037
wild-cluster bootstrap p (B = 1999)			0.035
pre-trend joint Wald (F , 3 and 45 DoF)	0.75		0.528
2018 event-study coefficient (outside selection window)	0.19	0.90	0.835
Placebo (largest civilian lines)	-0.95	0.25	0.001
<i>Leave-one-HS2-out jackknife</i>			
Drop HS 39 (1 line)	2.51	0.98	0.013
Drop HS 40 (1 line)	2.48	0.98	0.014
Drop HS 61 (2 lines)	2.67	1.00	0.009
Drop HS 84 (7 lines)	2.28	1.18	0.058
Drop HS 85 (13 lines)	1.36	1.06	0.207
Drop HS 90 (4 lines)	3.16	0.96	0.002
Drop HS 96 (1 line)	2.59	0.97	0.010
<i>Alternative surge-selection thresholds (both legs)</i>			
$1.5\times$ (40 lines)	1.59	0.90	0.080
$2.0\times$ (29 lines, baseline)	2.44	0.96	0.013
$2.5\times$ (23 lines)	3.12	1.08	0.005
$3.0\times$ (18 lines)	3.16	1.27	0.015
<i>Selection-rule-matched randomisation inference</i>			
Free permutation (year labels within HS6, rule re-applied): the rule selects a mean of 5.0 lines under H_0 (max 12 over 1,000 draws), so $P(\text{rule selects} \geq 29 \mid H_0) = 0.000$; the permuted-null γ has mean 2.79, s.d. 1.50, 95th percentile 5.37, and $P(\gamma^{\text{perm}} \geq 2.44) = 0.58$.			
Trend-preserving cyclic-shift null: mean 5.2 lines selected (max 19), $P(\geq 29 \mid H_0) = 0.000$; the cyclic-shift γ null (which preserves each line’s own trend and autocorrelation, unlike the free permutation) has mean 1.64, s.d. 1.20, and $P(\gamma^{\text{cyc}} \geq 2.44) = 0.29$ — narrower than the free-permutation null, as expected, but not overturning the conclusion that the magnitude is not cleanly separable from the selection.			
Priority list: a random 50-line basket drawn from all HS6 lines Kazakhstan exports to Russia has null γ mean 0.00, s.d. 0.97, and $P(\gamma^{\text{perm}} \geq 2.88) = 0.002$ — consistent with the priority-list coefficient being carried by its genuinely-treated overlap with the surge basket (Table 2), not itself an artefact of drawing 50 lines at random.			

Table D.1: Sensitivity of the value-capture headline to the retained margin m . Ratio = domestic value added per rerouted dollar, relative to per produced dollar ($m \bar{v}^{TT} / \bar{v}^M$, OECD ICIO).

m	0.06	0.10	0.14	0.19
VA per rerouted \$	0.047	0.079	0.110	0.150
Ratio to a produced \$	0.062	0.103	0.144	0.196
Retained, \$m (of \$479m)	23	38	53	72
m	0.25	0.32	0.34	0.49
VA per rerouted \$	0.197	0.252	0.268	0.386
Ratio to a produced \$	0.258	0.330	0.350	0.505
Retained, \$m (of \$479m)	94	121	128	185

Anchors: transport-only $m \approx 0.01$; wholesale-and-freight band 0.06–0.14; censored matched-cell gross margin 0.34; full trade-and-transport margin 0.49. The “corridor, not factory” reading (a rerouted dollar $\approx$ a tenth of a produced one) holds for m up to about 0.12; it weakens to “about a fifth” at $m \approx 0.19$, “about a third” at $m \approx 0.32$, and about a half at $m \approx 0.49$.

Table E.1: Surge-basket exports to Russia, USD million per year, and the annual break test.

	2018	2019	2020	2021	2022	2023	2024	2025	sup- F	p	jump yr
Kazakhstan (EAEU)	17	12	7	8	128	145	119	133	150.7	< 0.001	2022
Armenia (EAEU)	15	9	7	9	71	93	42	13	10.5	0.015	2022
Kyrgyz Republic (EAEU)	1	3	6	7	12	30	12	12	13.4	0.004	2019
Türkiye (non-EAEU)	53	55	47	58	119	207	98	73	11.9	0.007	2022
Georgia (non-EAEU)	0.2	0.7	0.4	0.7	1.0	1.1	1.7	0.7	6.9	0.075	2019

Kazakhstan row from Table 1. sup- F is a mean-shift test on a persistent series; we read it as evidence of a break, not a cardinal magnitude, and it does not itself date the break. “Largest jump” is the single year with the biggest year-on-year change in $\text{asinh}(\text{exports})$; it is a noisy diagnostic at these small, single-digit-to-low-double-digit levels, and for the Kyrgyz Republic and Georgia it lands on 2019 rather than 2022 despite both countries’ 2022 levels also rising over their 2018–2021 average (JIE round-2 review, Referee B N3). We read the sup- F rejection, not the single largest jump, as the evidence of a common disruption; the “same 2022 break” language elsewhere in the paper is best read as holding cleanly for Armenia and Türkiye, and as “elevated from 2022” rather than a dated break for the Kyrgyz Republic and Georgia, whose series are shorter and noisier. We attempted a Newey–West-corrected version of the sup- F test for each country; on these 8-year annual series the correction was not numerically stable (the bandwidth selection is itself unreliable at this sample size), unlike the ~ 70 -observation monthly series in Section 5, where it is decisive and leaves the confidence interval essentially unchanged.